Multilevel Main Effects Matrix Factor Model

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Abstract

We propose a multilevel main effects matrix factor model (MMEFM) to extract meaningful row and column effects from groups of matrix data. As a generalization of a multilevel matrix factor model, we provide rigorous definitions of MMEFM and identifications, together with iterative algorithms for the estimators of all global and local components in the main effects and common component, including global/local main effects and core ranks, with rates of convergence spelt out. An important feature of the model is that all main effects can be nonstationary and still be consistently estimated at each time point. At the same time, our estimated global common factors are alignment independent, and hence are truly global, and can be presented and given interpretations more easily. We propose a Bootstrap procedure for detecting the existence and identifying the groups of data that are driven by the global common factors. The power of the test, together with the accuracy of various estimators, are demonstrated in extensive simulation settings. Estimation performance is also compared to other state-of-the-art methods for matrix time series by simulations and a set of U.S. equity data. All numerical works are done using the R package `MMEFM`, which is available at `CRAN`.

Key words and phrases: Groups of panels; iterative projection; varying factor strength; Tucker decomposition; nonstationary main effects.

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1 Introduction

With rapid advances in data collection and storage technologies, large numbers of measurements are now routinely recorded over time in matrix form. Such matrix-valued time series arise, for example, as asset-by-indicator panels in finance, country-by-indicator panels in economics, and origin-by-destination traffic matrices. For such data, matrix factor models, which extend factor models for vector time series (Bai, 2003; Lam et al., 2011; Lam and Yao, 2012), provide a natural approach to dimension reduction by preserving the intrinsic row and column structure, rather than treating the vectorized observations through an unrestricted loading structure; see, among others, Wang et al. (2019), Chen et al. (2020), He et al. (2023a), Yu et al. (2022), Chen and Fan (2023) and He et al. (2023b). More recently, Lam and Cen (2025) introduce a main effects matrix factor model (MEFM) that explicitly separates a time-varying grand mean and row and column main effects from the interaction (common) component. This distinction allows different forms of systematic variation within a matrix to be modeled and interpreted separately.

In some applications, several related high-dimensional time series are observed, and their dimensions need not agree across groups. Such data may exhibit another layer of structure: some latent dynamics are shared across groups, while others are group-specific. This distinction has been well studied in the literature on multilevel factor models for vector time series; see, among others, Moench et al. (2013), Choi et al. (2018), Han (2021) and Choi et al. (2023). Related global–local factor structures have also been considered for higher-dimensional data. Lu et al. (2025) study a three-dimensional factor model with global and local factors, while He et al. (2025) consider common and cluster-specific factors for biclustering matrix-valued time series. More generally, Cen (2026) develops a multi-order tensor factor model that separates global and local components across multiple tensor time series. Most closely related to our setting, Zhang et al. (2025) propose a multilevel matrix factor model (MFM) with global and local bilinear components. A global matrix factor process is shared among groups through group-specific row and column loadings, while local matrix factors capture group-specific variation.

The two forms of structure, multilevel variation across groups and additive main effects within each matrix, are complementary and can arise simultaneously in grouped matrix-valued time series. As pointed out by Lam and Cen (2025), additive main effects can be absorbed into an enlarged bilinear representation, but treating such a representation as unstructured obscures their distinct interpretation and rank structure. We therefore introduce a **M**ultilevel **M**ain **E**ffects matrix **F**actor **M**odel (MMEFM), which extends the MFM of Zhang et al. (2025) by incorporating time-varying grand means and row and column main effects, each with global and local components. While Lam and Cen (2025) discuss low-rank factor representations for the main effects, we develop such structures in a multilevel setting by decomposing the row and column main effects into global and local components, respectively. This exploits cross-group information and

provides a parsimonious representation while retaining group-specific dynamics. At the model level, the MFM of Zhang et al. (2025) is a restricted case of the MMEFM, so that the latter provides a more general representation by allowing global and local main-effect structures; see Remark 2.

Using pre-averaging ideas from Chen and Lam (2024), our estimation procedure utilizes cross-group information for the global main effects as well as the global common factors in the common component through projected cross-group moments. Local components in each data group are then estimated and removed from the data, with global factor processes still expressed under different row and column transformations. We therefore align and pool them through a nearest-Kronecker approximation to obtain a global common factor series $\widehat{\mathbf{G}}_t$, followed by a refitting the group-specific global loadings. This leads to one of our contributions distinguished from Zhang et al. (2025), that unlike their group-specific representations $\widehat{\mathbf{S}}_{mt}$ of the global common factors which are still group-dependent, our global main effects and common factor series are truly common across groups. They can then be easier visualized and compared with other estimated group-specific components to provide further insight.

We further establish convergence rates for the proposed estimators under a class of piecewise-stationary linear factor processes. In particular, the class $\mathcal{G}_T$ accommodates both short- and long-memory dependence as well as changes in dynamics; see Section 4 for details. Within the same setting, we also provide rank estimators of all loading matrices in Section 3.4, with theoretical results presented in Section 4.

Another important contribution is to determine the presence of a global common component. Imposing such a component without first assessing its presence may give a false impression of shared dynamics when no cross-group common factor exists. We construct group-level statistics and show that the order of the statistics for groups driven by the global components are larger than those for the remaining groups. Building on this separation, we develop a practical existence test for the global common components together with a screening rule for identifying the groups that share them.

The remainder of the paper is organized as follows. Section 2 introduces notation. Section 3 presents the MMEFM and its estimation. Section 4 introduces the assumptions for the MMEFM and establishes its theoretical properties, while Section 5 presents the practical detection and screening procedure for global common factors. Section 6 reports simulation results, including comparisons with other state-of-the-art matrix factor models, and an empirical analysis of the U.S. stock market. Proofs and additional results are included in the Supplementary Materials.

2 Notations

Throughout this paper, we use a letter, bold lower-case letter, and bold capital letter, i.e., a or A, $\mathbf{a}$, and $\mathbf{A}$, to denote a scalar, a vector, and a matrix respectively. We also use $a_i, A_{i,j}, \mathbf{A}_{i\cdot}, \mathbf{A}_{\cdot i}$ to denote, respectively,

the ith element of $\mathbf{a}$, the (i,j)th element of $\mathbf{A}$, the ith row vector (as a column vector) of $\mathbf{A}$, and the ith column vector of $\mathbf{A}$. For any $a, b \in \mathbb{R}$, we write $a \vee b := \max\{a, b\}$ and $a \wedge b := \min\{a, b\}$. We use $a \asymp b$ to denote $a = O(b)$ and $b = O(a)$, while $a \asymp_P b$ to denote $a = O_P(b)$ and $b = O_P(a)$. We also use $a \gtrsim b$ and $a \lesssim b$ to denote $b = O(a)$ and $a = O(b)$ respectively, and $\gtrsim_P$ and $\lesssim_P$ their respective asymptotic probabilistic versions. We use $\circ$ to denote the Hadamard product and $\otimes$ the Kronecker product.

Given a positive integer m, define $[m] := \{1, 2, \ldots, m\}$. The vector $\mathbf{1}_m$ denotes a vector of ones with length m. The ith largest eigenvalue of a matrix $\mathbf{A}$ is denoted by $\lambda_i(\mathbf{A})$. The notation $\mathbf{A} \succcurlyeq 0$ (resp. $\mathbf{A} \succ 0$) means that $\mathbf{A}$ is positive semi-definite (resp. positive definite). We use $\mathbf{A}'$ to denote the transpose of $\mathbf{A}$, and $\mathrm{diag}(\mathbf{A})$ to denote a diagonal matrix with the diagonal elements of $\mathbf{A}$, while $\mathrm{diag}(\{a_1, \ldots, a_n\})$ represents the diagonal matrix with $\{a_1, \ldots, a_n\}$ on the diagonal. The matrix $\mathbf{A}^\perp$ has orthonormal columns spanning the orthogonal complement of the column space of $\mathbf{A}$, so that $\mathbf{A}'\mathbf{A}^\perp = \mathbf{0}$ and $\mathbf{A}^{\perp\prime}\mathbf{A}^\perp = \mathbf{I}$. We use $\mathbf{A}^{\otimes 2}$ to denote $\mathbf{A}\mathbf{A}'$.

Some norm notations. For a given set, we denote by $|\cdot|$ its cardinality. We use $\|\cdot\|$ to denote the spectral norm of a matrix or the L_2 norm of a vector, and $\|\cdot\|_F$ to denote the Frobenius norm of a matrix. We use $\|\cdot\|_{\max}$ to denote the maximum absolute value of the elements in a vector or a matrix. The notations $\|\cdot\|_1$ and $\|\cdot\|_\infty$ denote the L_1 and L_∞-norm of a matrix respectively, defined by $\|\mathbf{A}\|_1 := \max_j \sum_i |(\mathbf{A})_{ij}|$ and $\|\mathbf{A}\|_\infty := \max_i \sum_j |(\mathbf{A})_{ij}|$. Without loss of generality, we always assume the eigenvalues of a matrix are arranged by descending orders, and so are their corresponding eigenvectors.

3 Model and Estimation

We propose the MMEFM as follows. For each data group $m \in [M]$ and $t \in [T]$,

$$
\begin{aligned}
\mathbf{Y}_t^{(m)} &= (\mu_t + \mu_t^{(m)})\mathbf{1}_{p_m}\mathbf{1}_{q_m}' + [(\mathbf{A}_{1m}\boldsymbol{\alpha}_t) + (\mathbf{A}_{2m}\boldsymbol{\alpha}_t^{(m)})]\mathbf{1}_{q_m}' + \mathbf{1}_{p_m}[(\mathbf{B}_{1m}\boldsymbol{\beta}_t) + (\mathbf{B}_{2m}\boldsymbol{\beta}_t^{(m)})]' \\
&\quad + \mathbf{Q}_m\mathbf{G}_t\mathbf{J}_m' + \mathbf{R}_m\mathbf{F}_t^{(m)}\mathbf{C}_m' + \mathbf{E}_t^{(m)} \\
&= \mu_t\mathbf{1}_{p_m}\mathbf{1}_{q_m}' + (\mathbf{A}_{1m}\boldsymbol{\alpha}_t)\mathbf{1}_{q_m}' + \mathbf{1}_{p_m}(\mathbf{B}_{1m}\boldsymbol{\beta}_t)' + \mathbf{Q}_m\mathbf{G}_t\mathbf{J}_m' \\
&\quad + \mu_t^{(m)}\mathbf{1}_{p_m}\mathbf{1}_{q_m}' + (\mathbf{A}_{2m}\boldsymbol{\alpha}_t^{(m)})\mathbf{1}_{q_m}' + \mathbf{1}_{p_m}(\mathbf{B}_{2m}\boldsymbol{\beta}_t^{(m)})' + \mathbf{R}_m\mathbf{F}_t^{(m)}\mathbf{C}_m' + \mathbf{E}_t^{(m)},
\end{aligned} \tag{3.1}
$$

where $\mathbf{Y}_t^{(m)}$ is the mth group observed matrix of dimension $p_m \times q_m$ at time t, μ_t, $\boldsymbol{\alpha}_t$, $\boldsymbol{\beta}_t$, $\mu_t^{(m)}$, $\boldsymbol{\alpha}_t^{(m)}$ and $\boldsymbol{\beta}_t^{(m)}$ are the global grand mean, row main effects, column main effects, and their corresponding local versions at group m respectively at time t. The matrices $\mathbf{A}_{1m} \in \mathbb{R}^{p_m \times r_1}$ (resp. $\mathbf{A}_{2m} \in \mathbb{R}^{p_m \times r_2^{(m)}}$) and $\mathbf{B}_{1m} \in \mathbb{R}^{q_m \times \ell_1}$ (resp. $\mathbf{B}_{2m} \in \mathbb{R}^{q_m \times \ell_2^{(m)}}$) are the global (resp. local) row main effect and the global (resp. local) column main effect loading matrix respectively at group m. The matrices $\mathbf{Q}_m \in \mathbb{R}^{p_m \times k_r}$, $\mathbf{J}_m \in \mathbb{R}^{q_m \times k_c}$, $\mathbf{R}_m \in \mathbb{R}^{p_m \times k_r^{(m)}}$ and $\mathbf{C}_m \in \mathbb{R}^{q_m \times k_c^{(m)}}$ are the global row, global column, local row and local

column factor loading matrix respectively at group m. Finally, $\mathbf{G}_t$ and $\mathbf{F}_t^{(m)}$ are the global factor and local factor matrices (at group m) respectively.

The upper equality of model (3.1) lists the grand means and the main effects first before the factor components, whereas the lower equality lists the global components first before the local components. This model can be considered a generalization of the one in Zhang et al. (2025), with global and local main effects included in our model. It is clear that the global time series $\{\boldsymbol{\alpha}_t\}$, $\{\boldsymbol{\beta}_t\}$ and $\{\mathbf{G}_t\}$ are all group-independent, whereas all other components of the model are dependent on the group m. It is also clear that at time t, the grand mean, row and column main effects for the group m data are given respectively by

$$c_t^{(m)} := \mu_t + \mu_t^{(m)}, \quad \mathbf{a}_t^{(m)} := \mathbf{A}_{1m}\boldsymbol{\alpha}_t + \mathbf{A}_{2m}\boldsymbol{\alpha}_t^{(m)}, \quad \mathbf{b}_t^{(m)} := \mathbf{B}_{1m}\boldsymbol{\beta}_t + \mathbf{B}_{2m}\boldsymbol{\beta}_t^{(m)}, \tag{3.2}$$

which are all an affine transformation of the global and local grand means/main effects and are naturally unidentifiable without further restrictions. The same applies to the global and local common components $\mathbf{Q}_m\mathbf{G}_t\mathbf{J}_m'$ and $\mathbf{R}_m\mathbf{F}_t^{(m)}\mathbf{C}_m'$, which are themselves unidentifiable because they are the product of unknown matrices.

With (3.2), model (3.1) can also be written as, for each $t \in [T]$,

$$\mathbf{Y}_t^{(m)} = c_t^{(m)}\mathbf{1}_{p_m}\mathbf{1}_{q_m}' + \mathbf{a}_t^{(m)}\mathbf{1}_{q_m}' + \mathbf{1}_{p_m}\mathbf{b}_t^{(m)\prime} + (\mathbf{Q}_m, \mathbf{R}_m)\begin{pmatrix} \mathbf{G}_t & \mathbf{0} \\ \mathbf{0} & \mathbf{F}_t^{(m)} \end{pmatrix}\begin{pmatrix} \mathbf{J}_m' \\ \mathbf{C}_m' \end{pmatrix} + \mathbf{E}_t^{(m)}, \quad m \in [M],$$

which is a main effect matrix factor model in Lam and Cen (2025) for each group m, with a special block diagonal structure for the factor matrix. If we ignore this structure, the total number of factor series to be estimated is then $\sum_{m=1}^{M}(k_r + k_r^{(m)})(k_c + k_c^{(m)})$, which is much larger than $k_r k_c + \sum_{m=1}^{M} k_r^{(m)} k_c^{(m)}$, the number of factor series in model (3.1). More importantly, the interplay between these groups of time series and thus their interpretation would be lost.

Remark 1 *Model (3.1) has the main effects split into "global" and "local", apparently increasing the number of parameters and main effect series to estimate. However, as is evident in Section 3.3, the split is mainly there to promote interpretation rather than to increase the difficulty of estimation, since* $c_t^{(m)}$, $\mathbf{a}_t^{(m)}$ *and* $\mathbf{b}_t^{(m)}$ *can all be estimated consistently for each* $t \in [T]$ *from Theorem 2.*

3.1 Examples

Consider the case where each data group represents an industry sector. For instance, $M = 11$ and each data matrix from group m, $\mathbf{Y}_t^{(m)}$, records data at time t from one of the eleven sectors classified by the Global Industry Classification Standard (GICS). Within each group, we may consider countries from the

G20 as the rows of $\mathbf{Y}_t^{(m)}$, and various economic indicators of each industry/sub-industry related to the sector at time t as the columns. Hence in this example we have $p_m = 20$ countries, but q_m can be different for each $m \in [11]$, depending on what economic indicators and what industry/sub-industry classifications we go for in each sector.

As for interpretations, the term $(\mathbf{A}_{1m}\boldsymbol{\alpha}_t)\mathbf{1}'_{q_m}$ from model (3.1) represents the country main effects which are *the same* across different economic indicators/ industries/sub-industries. The same vector $\boldsymbol{\alpha}_t$ continues to drive the country main effects across different sectors. Any genuine variation for the country main effects arising from differences of sectors are captured by the terms $(\mathbf{A}_{2m}\boldsymbol{\alpha}_t^{(m)})\mathbf{1}'_{q_m}$ for $m \in [11]$. Reversing the roles of country and economic indicators/industries/sub-industries while replacing $\boldsymbol{\alpha}_t$ with $\boldsymbol{\beta}_t$ above give interpretations of $\mathbf{1}_{p_m}(\mathbf{B}_{1m}\boldsymbol{\beta}_t)'$ and $\mathbf{1}_{p_m}(\mathbf{B}_{2m}\boldsymbol{\beta}_t^{(m)})'$ respectively. Finally, the term $\mathbf{Q}_m\mathbf{G}_t\mathbf{J}'_m$ represents any *global* interaction effects remaining after accounting for all the global and local main effects, which are all driven by some global factor time series in $\mathbf{G}_t$. Any genuine differences in interaction effects across sectors are captured by the terms $\mathbf{R}_m\mathbf{F}_t^{(m)}\mathbf{C}'_m$.

Another example would be the financial data set used in Zhang et al. (2025). Same as the example described above, each data group still represents an industry sector. However, each row now represents a stock within the sector, and there are $p_m = 20$ stocks included for each sector. For the columns, financial data like return, volume of trade and turnover rate are included, so that q_m are the same for each m.

3.2 Identification

We first present (IC1) for identifying the main effects from the common component and (IC2) for identifying the global effects from the local ones. Condition (IC3) is needed for separating the loading matrices from the time series, so that both can be identified.

(IC1) (Main effects v.s. common component) For $m \in [M]$ and $j = 1, 2$, we have

$$\mathbf{1}'_{p_m}\mathbf{A}_{jm} = \mathbf{0},\ \mathbf{1}'_{q_m}\mathbf{B}_{jm} = \mathbf{0},\ \mathbf{1}'_{p_m}\mathbf{Q}_m = \mathbf{0},\ \mathbf{1}'_{q_m}\mathbf{J}_m = \mathbf{0},\ \mathbf{1}'_{p_m}\mathbf{R}_m = \mathbf{0},\ \mathbf{1}'_{q_m}\mathbf{C}_m = \mathbf{0}.$$

(IC2) (Global v.s. local) For each $m \in [M]$, the series $\{\mathbf{G}_t\}$ is uncorrelated with $\{\mathbf{F}_t^{(m)}\}$. The series $\{\boldsymbol{\alpha}_t^{(m)}\}$, $\{\boldsymbol{\beta}_t^{(m)}\}$ and $\{\mathbf{F}_t^{(m)}\}$ are uncorrelated to $\{\boldsymbol{\alpha}_t^{(n)}\}$, $\{\boldsymbol{\beta}_t^{(n)}\}$ and $\{\mathbf{F}_t^{(n)}\}$ respectively for all $m \neq n$. Also $\sum_{m=1}^{M}\mu_t^{(m)} = 0$, $\sum_{t=1}^{T}\boldsymbol{\alpha}_t\boldsymbol{\alpha}_t^{(m)'} = \mathbf{0}$ and $\sum_{t=1}^{T}\boldsymbol{\beta}_t\boldsymbol{\beta}_t^{(m)'} = \mathbf{0}$ for each $m \in [M]$. The elements of error series $\{\mathbf{E}_t^{(m)}\}$ are uncorrelated with those of $\{\mathbf{E}_t^{(n)}\}$ for $m \neq n$, and are also uncorrelated with those of $\{\boldsymbol{\alpha}_t\}$, $\{\boldsymbol{\beta}_t\}$, $\{\mathbf{G}_t\}$, $\{\boldsymbol{\alpha}_t^{(s)}\}$, $\{\boldsymbol{\beta}_t^{(s)}\}$ and $\{\mathbf{F}_t^{(s)}\}$ for all $s \in [M]$.

(IC3) (Separation of loading matrices and time series) All loading matrices involved in model (3.1) are of full rank; of full rank are also the matrices $\mathbb{E}(\boldsymbol{\alpha}_t^{\otimes 2})$, $\mathbb{E}(\boldsymbol{\beta}_t^{\otimes 2})$, $\mathbb{E}(\boldsymbol{\alpha}_t^{(m)\otimes 2})$, $\mathbb{E}(\boldsymbol{\beta}_t^{(m)\otimes 2})$, $\mathbb{E}(\mathbf{vec}(\mathbf{G}_t)^{\otimes 2})$,

$\mathbb{E}(\mathbf{vec}(\mathbf{F}_t^{(m)})^{\otimes 2})$, $\mathbf{A}_{1m}^{\perp\prime}\mathbf{A}_{2m}$, $\mathbf{B}_{1m}^{\perp\prime}\mathbf{B}_{2m}$, $\mathbf{Q}_m^{\perp\prime}\mathbf{R}_m$ and $\mathbf{J}_m^{\perp\prime}\mathbf{C}_m$.

The assumptions $\sum_{t=1}^T \boldsymbol{\alpha}_t\boldsymbol{\alpha}_t^{(m)\prime} = \mathbf{0}$ and $\sum_{t=1}^T \boldsymbol{\beta}_t\boldsymbol{\beta}_t^{(m)\prime} = \mathbf{0}$ in (IC2) are not restrictive at all and are only for convenience in relevant proofs. For instance,

$$\begin{aligned}\mathbf{A}_{1m}\boldsymbol{\alpha}_t + \mathbf{A}_{2m}\boldsymbol{\alpha}_t^{(m)} &= \left\{\mathbf{A}_{1m} + \mathbf{A}_{2m}\left(\sum_{s=1}^T \boldsymbol{\alpha}_s^{(m)}\boldsymbol{\alpha}_s'\right)\left(\sum_{s=1}^T \boldsymbol{\alpha}_s\boldsymbol{\alpha}_s'\right)^{-1}\right\}\boldsymbol{\alpha}_t \\ &\quad + \mathbf{A}_{2m}\left\{\boldsymbol{\alpha}_t^{(m)} - \left(\sum_{s=1}^T \boldsymbol{\alpha}_s^{(m)}\boldsymbol{\alpha}_s'\right)\left(\sum_{s=1}^T \boldsymbol{\alpha}_s\boldsymbol{\alpha}_s'\right)^{-1}\boldsymbol{\alpha}_t\right\} =: \widetilde{\mathbf{A}}_{1m}\boldsymbol{\alpha}_t + \mathbf{A}_{2m}\widetilde{\boldsymbol{\alpha}}_t^{(m)},\end{aligned}$$

now with $\sum_{t=1}^T \boldsymbol{\alpha}_t\widetilde{\boldsymbol{\alpha}}_t^{(m)\prime} = \mathbf{0}$. The (asymptotic) invertibility of $\sum_{t=1}^T \boldsymbol{\alpha}_t\boldsymbol{\alpha}_t'$ comes from Assumption (IC3), and (S1) in Section 4.1. Meanwhile, if $\{\boldsymbol{\alpha}_t\}$ is uncorrelated with $\{\boldsymbol{\alpha}_t^{(m)}\}$ for $m \in [M]$, then elements in $T^{-1}\sum_{t=1}^T \widetilde{\boldsymbol{\alpha}}_t^{(m)}\widetilde{\boldsymbol{\alpha}}_t^{(n)\prime}$ are going to 0 at the same rate as those in $T^{-1}\sum_{t=1}^T \boldsymbol{\alpha}_t^{(m)}\boldsymbol{\alpha}_t^{(n)\prime}$ for $m \neq n$, rendering $\{\widetilde{\boldsymbol{\alpha}}_t^{(m)}\}$ and $\{\widetilde{\boldsymbol{\alpha}}_t^{(n)}\}$ asymptotically uncorrelated for $m \neq n$. For simplicity, we do not engage in asymptotic identification in the proof of Theorem 1 and only use (IC2) as above for exact identification.

Theorem 1 (Identification) *Under (IC1), (IC2) and (IC3), all main effects, loading matrices, global and local means and common factors in model (3.1) are identified.*

Remark 2 *Following a demonstration in Lam and Cen (2025), using the notations in (3.2), model (3.1) can be written as a multilevel matrix FM as follows:*

$$\mathbf{Y}_t^{(m)} = \widetilde{\mathbf{Q}}_m\widetilde{\mathbf{W}}_t^{(m)}\widetilde{\mathbf{J}}_m' + \mathbf{E}_t^{(m)}, \quad \textit{where}$$

$$\widetilde{\mathbf{Q}}_m := (\mathbf{I}_{p_m}, \mathbf{Q}_m, \mathbf{R}_m, \mathbf{1}_{p_m}), \quad \widetilde{\mathbf{J}}_m := (\mathbf{1}_{q_m}, \mathbf{J}_m, \mathbf{C}_m, \mathbf{I}_{q_m}), \quad \widetilde{\mathbf{W}}_t^{(m)} := \begin{pmatrix} \mathbf{a}_t^{(m)} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{G}_t & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{F}_t^{(m)} & \mathbf{0} \\ c_t^{(m)} & \mathbf{0} & \mathbf{0} & \mathbf{b}_t^{(m)\prime} \end{pmatrix}.$$

Ignoring the above structure means that we are estimating $(p_m + k_r + k_r^{(m)} + 1)(q_m + k_c + k_c^{(m)} + 1)$ time series in $\widetilde{\mathbf{W}}_t^{(m)}$, which tends to overfit the data because $p_m q_m$ is usually too large relative to T.

On the other hand, consider a multilevel matrix FM given by $\acute{\mathbf{Y}}_t^{(m)} = \acute{\mathbf{C}}_t + \acute{\mathbf{E}}_t$ for $m \in [M]$, where $\acute{\mathbf{C}}_t := \mathbf{Q}_m\mathbf{G}_t\mathbf{J}_m' + \mathbf{R}_m\mathbf{F}_t^{(m)}\mathbf{C}_m'$. Defining $\mathbf{M}_n := \mathbf{I}_n - n^{-1}\mathbf{1}_n\mathbf{1}_n'$, we construct

$$c_t^{(m)} := (p_m q_m)^{-1}\mathbf{1}_{p_m}'\acute{\mathbf{C}}_t\mathbf{1}_{q_m}, \quad \mathbf{a}_t^{(m)} := q_m^{-1}\mathbf{M}_{p_m}\acute{\mathbf{C}}_t\mathbf{1}_{q_m}, \quad \mathbf{b}_t^{(m)} := p_m^{-1}\mathbf{M}_{q_m}\acute{\mathbf{C}}_t'\mathbf{1}_{p_m}.$$

Then $\mathbf{Y}_t^{(m)}$ *can be expressed as*

$$\mathbf{Y}_t^{(m)} = c_t^{(m)}\mathbf{1}_{p_m}\mathbf{1}_{q_m}' + \mathbf{a}_t^{(m)}\mathbf{1}_{q_m}' + \mathbf{1}_{p_m}\mathbf{b}_t^{(m)'} + (\mathbf{M}_{p_m}\mathbf{Q}_m)\mathbf{G}_t(\mathbf{M}_{q_m}\mathbf{J}_m)' + (\mathbf{M}_{p_m}\mathbf{R}_m)\mathbf{F}_t^{(m)}(\mathbf{M}_{q_m}\mathbf{C}_m)' + \acute{\mathbf{E}}_t,$$

which is a multilevel MEFM in (3.1), with grand means and main effects given by

$$\mu_t = (p_m q_m)^{-1}\mathbf{1}_{p_m}'\mathbf{Q}_m\mathbf{G}_t\mathbf{J}_m'\mathbf{1}_{q_m},\ \mu_t^{(m)} = (p_m q_m)^{-1}\mathbf{1}_{p_m}'\mathbf{R}_m\mathbf{F}_t^{(m)}\mathbf{C}_m'\mathbf{1}_{q_m},\ \boldsymbol{\alpha}_t = \mathbf{G}_t\mathbf{J}_m'\mathbf{1}_{q_m},$$
$$\boldsymbol{\beta}_t = \mathbf{G}_t'\mathbf{Q}_m'\mathbf{1}_{p_m},\ \boldsymbol{\alpha}_t^{(m)} = \mathbf{F}_t^{(m)}\mathbf{C}_m'\mathbf{1}_{q_m},\ \boldsymbol{\beta}_t^{(m)} = \mathbf{F}_t^{(m)'}\mathbf{R}_m'\mathbf{1}_{p_m},$$

and the associated main effect loading matrices for $\boldsymbol{\alpha}_t$, $\boldsymbol{\beta}_t$, $\boldsymbol{\alpha}_t^{(m)}$ *and* $\boldsymbol{\beta}_t^{(m)}$ *given respectively by*

$$\mathbf{A}_{1m} = q_m^{-1}\mathbf{M}_{p_m}\mathbf{Q}_m,\ \ \mathbf{B}_{1m} = p_m^{-1}\mathbf{M}_{q_m}\mathbf{J}_m,\ \ \mathbf{A}_{2m} = q_m^{-1}\mathbf{M}_{p_m}\mathbf{R}_m,\ \ \mathbf{B}_{2m} = p_m^{-1}\mathbf{M}_{q_m}\mathbf{C}_m.$$

Two important observations are that all factor loading matrices are now one rank less because of the multiplication of $\mathbf{M}_{p_m}$ *and* $\mathbf{M}_{q_m}$*, and* $\boldsymbol{\alpha}_t^{(m)}$ *and* $\boldsymbol{\beta}_t^{(m)}$ *are now of length* $k_r^{(m)}$ *and* $k_c^{(m)}$ *rather than being* $r_2^{(m)}$ *and* $\ell_2^{(m)}$ *respectively in model (3.1). Since* $r_2^{(m)}$ *and* $\ell_2^{(m)}$ *can be larger than* $k_r^{(m)}$ *and* $k_c^{(m)}$*, a multilevel MEFM is certainly more flexible than a multilevel matrix FM.*

3.3 Estimation

The proof of Theorem 1 is instrumental to the estimators to be presented below. We first present the estimators for the grand mean and main effects as a whole. Then we present all global and local estimators for the main effects, before all common component estimators.

Hereafter, whenever an estimated loading matrix is formed from the first k eigenvectors of a symmetric matrix $\mathbf{V}$, these eigenvectors are chosen to be orthonormal and to correspond to the k largest eigenvalues of $\mathbf{V}$.

1. (Grand mean and main effects) The estimators of $c_t^{(m)}$, $\mathbf{a}_t^{(m)}$ and $\mathbf{b}_t^{(m)}$ are given respectively by

$$\widehat{c}_t^{(m)} := (p_m q_m)^{-1}\mathbf{1}_{p_m}'\mathbf{Y}_t^{(m)}\mathbf{1}_{q_m},\quad \widehat{\mathbf{a}}_t^{(m)} := q_m^{-1}\mathbf{M}_{p_m}\mathbf{Y}_t^{(m)}\mathbf{1}_{q_m},\quad \widehat{\mathbf{b}}_t^{(m)} := p_m^{-1}\mathbf{M}_{q_m}\mathbf{Y}_t^{(m)'}\mathbf{1}_{p_m}.$$

We estimate the global and local grand means by respectively

$$\widehat{\mu}_t := M^{-1}\sum_{m=1}^{M}\widehat{c}_t^{(m)},\quad \widehat{\mu}_t^{(m)} := \widehat{c}_t^{(m)} - \widehat{\mu}_t.$$

2. (Global and local main effects) Define

$$\widehat{\mathbf{\Omega}}_{A_1,m} := (M-1)^{-1} \sum_{n=1;n\neq m}^{M} \left\{ (Tq_mq_n)^{-1} \mathbf{M}_{p_m} \sum_{t=1}^{T} \mathbf{Y}_t^{(m)} \mathbf{1}_{q_m} \mathbf{1}'_{q_n} \mathbf{Y}_t^{(n)'} \mathbf{M}_{p_n} \right\}^{\otimes 2}, \tag{3.3}$$

$$\widehat{\mathbf{\Omega}}_{B_1,m} := (M-1)^{-1} \sum_{n=1;n\neq m}^{M} \left\{ (Tp_mp_n)^{-1} \mathbf{M}_{q_m} \sum_{t=1}^{T} \mathbf{Y}_t^{(m)'} \mathbf{1}_{p_m} \mathbf{1}'_{p_n} \mathbf{Y}_t^{(n)} \mathbf{M}_{q_n} \right\}^{\otimes 2}. \tag{3.4}$$

We set $\widehat{\mathbf{A}}_{1m}$ and $\widehat{\mathbf{B}}_{1m}$ respectively as the first r_1 eigenvectors of $\widehat{\mathbf{\Omega}}_{A_1,m}$, and the first ℓ_1 eigenvectors of $\widehat{\mathbf{\Omega}}_{B_1,m}$. We take $\widehat{\mathbf{A}}_{2m}$ and $\widehat{\mathbf{B}}_{2m}$ respectively as the first $r_2^{(m)}$ and $\ell_2^{(m)}$ eigenvectors of

$$\widehat{\mathbf{\Omega}}_{A_2,m} := \left\{ T^{-1} \sum_{t=1}^{T} \widehat{\mathbf{a}}_t^{(m)\otimes 2} \widehat{\mathbf{A}}_{1m}^{\perp} \right\}^{\otimes 2} \text{ and } \widehat{\mathbf{\Omega}}_{B_2,m} := \left\{ T^{-1} \sum_{t=1}^{T} \widehat{\mathbf{b}}_t^{(m)\otimes 2} \widehat{\mathbf{B}}_{1m}^{\perp} \right\}^{\otimes 2}. \tag{3.5}$$

The local main effect time series are estimated as

$$\begin{aligned} \widehat{\boldsymbol{\alpha}}_t^{(m)} &:= q_m^{-1} (\widehat{\mathbf{A}}'_{2m} \widehat{\mathbf{A}}_{1m}^{\perp} \widehat{\mathbf{A}}_{1m}^{\perp'} \widehat{\mathbf{A}}_{2m})^{-1} \widehat{\mathbf{A}}'_{2m} \widehat{\mathbf{A}}_{1m}^{\perp} \widehat{\mathbf{A}}_{1m}^{\perp'} \mathbf{M}_{p_m} \mathbf{Y}_t^{(m)} \mathbf{1}_{q_m}, \\ \widehat{\boldsymbol{\beta}}_t^{(m)} &:= p_m^{-1} (\widehat{\mathbf{B}}'_{2m} \widehat{\mathbf{B}}_{1m}^{\perp} \widehat{\mathbf{B}}_{1m}^{\perp'} \widehat{\mathbf{B}}_{2m})^{-1} \widehat{\mathbf{B}}'_{2m} \widehat{\mathbf{B}}_{1m}^{\perp} \widehat{\mathbf{B}}_{1m}^{\perp'} \mathbf{M}_{q_m} \mathbf{Y}_t^{(m)'} \mathbf{1}_{p_m}. \end{aligned} \tag{3.6}$$

Define

$$\begin{aligned} \widehat{\mathbf{Z}}_{\alpha,m} &:= \widehat{\mathbf{A}}'_{1m} (\widehat{\mathbf{y}}_1^{(m)}, \ldots, \widehat{\mathbf{y}}_T^{(m)}), \quad \text{where} \quad \widehat{\mathbf{y}}_t^{(m)} := q_m^{-1} \mathbf{M}_{p_m} \mathbf{Y}_t^{(m)} \mathbf{1}_{q_m} - \widehat{\mathbf{A}}_{2m} \widehat{\boldsymbol{\alpha}}_t^{(m)}, \\ \widehat{\mathbf{Z}}_{\beta,m} &:= \widehat{\mathbf{B}}'_{1m} (\widehat{\mathbf{w}}_1^{(m)}, \ldots, \widehat{\mathbf{w}}_T^{(m)}), \quad \text{where} \quad \widehat{\mathbf{w}}_t^{(m)} := p_m^{-1} \mathbf{M}_{q_m} \mathbf{Y}_t^{(m)'} \mathbf{1}_{p_m} - \widehat{\mathbf{B}}_{2m} \widehat{\boldsymbol{\beta}}_t^{(m)}. \end{aligned}$$

We estimate $\widehat{\mathbf{A}}_\alpha := (\widehat{\boldsymbol{\alpha}}_1, \ldots, \widehat{\boldsymbol{\alpha}}_T)$ and $\widehat{\mathbf{B}}_\beta := (\widehat{\boldsymbol{\beta}}_1, \ldots, \widehat{\boldsymbol{\beta}}_T)$ as $\sqrt{T}$ times the matrices whose rows are, respectively, the first r_1 and ℓ_1 eigenvectors of

$$M^{-1} \sum_{m=1}^{M} \widehat{\mathbf{Z}}'_{\alpha,m} (\widehat{\mathbf{Z}}_{\alpha,m} \widehat{\mathbf{Z}}'_{\alpha,m})^{-1} \widehat{\mathbf{Z}}_{\alpha,m} \quad \text{and} \quad M^{-1} \sum_{m=1}^{M} \widehat{\mathbf{Z}}'_{\beta,m} (\widehat{\mathbf{Z}}_{\beta,m} \widehat{\mathbf{Z}}'_{\beta,m})^{-1} \widehat{\mathbf{Z}}_{\beta,m}.$$

We then re-estimate $\mathbf{A}_{1m}$ and $\mathbf{B}_{1m}$ respectively as

$$\widetilde{\mathbf{A}}_{1m} := T^{-1} (\widehat{\mathbf{y}}_1^{(m)}, \ldots, \widehat{\mathbf{y}}_T^{(m)}) \widehat{\mathbf{A}}'_\alpha \quad \text{and} \quad \widetilde{\mathbf{B}}_{1m} := T^{-1} (\widehat{\mathbf{w}}_1^{(m)}, \ldots, \widehat{\mathbf{w}}_T^{(m)}) \widehat{\mathbf{B}}'_\beta. \tag{3.7}$$

3. (Common component) Define

$$\breve{\mathbf{Y}}_t^{(m)} := \mathbf{Y}_t^{(m)} - \widehat{c}_t^{(m)} \mathbf{1}_{p_m} \mathbf{1}'_{q_m} - \widehat{\mathbf{a}}_t^{(m)} \mathbf{1}'_{q_m} - \mathbf{1}_{p_m} \widehat{\mathbf{b}}_t^{(m)'}.$$

Using the pre-averaging idea in Chen and Lam (2024), we find from a family of randomly generated

directions the one that aligns the most with the direction of the strongest factors from $\mathbf{Q}_m$, say $\mathbf{v}_a^{(m)}$. Do the same for $\mathbf{J}_m$, say $\mathbf{v}_b^{(m)}$. See Section A for all the details of this procedure.

Let $\mathbf{v}_a^{(m)}$ and $\mathbf{v}_b^{(m)}$ be standardized as unit vectors. Define

$$\mathbf{X}_{Q,m} := T^{-2}(M-1)^{-1} \sum_{n=1;n\neq m}^{M} \left\{ \sum_{t=1}^{T} \breve{\mathbf{Y}}_t^{(m)} \mathbf{v}_b^{(m)} \mathbf{v}_b^{(n)\prime} \breve{\mathbf{Y}}_t^{(n)\prime} \right\}^{\otimes 2}, \tag{3.8}$$

$$\mathbf{X}_{J,m} := T^{-2}(M-1)^{-1} \sum_{n=1;n\neq m}^{M} \left\{ \sum_{t=1}^{T} \breve{\mathbf{Y}}_t^{(m)\prime} \mathbf{v}_a^{(m)} \mathbf{v}_a^{(n)\prime} \breve{\mathbf{Y}}_t^{(n)} \right\}^{\otimes 2}. \tag{3.9}$$

We take $\widehat{\mathbf{Q}}_m$ as the first k_r eigenvectors of $\mathbf{X}_{Q,m}$, and $\widehat{\mathbf{J}}_m$ as the first k_c eigenvectors of $\mathbf{X}_{J,m}$.

We then define

$$\mathbf{X}_{R,m} := T^{-1} \sum_{t=1}^{T} \{ \breve{\mathbf{Y}}_t^{(m)} \widehat{\mathbf{J}}_m^{\perp} \}^{\otimes 2}, \quad \mathbf{X}_{C,m} := T^{-1} \sum_{t=1}^{T} \{ \breve{\mathbf{Y}}_t^{(m)\prime} \widehat{\mathbf{Q}}_m^{\perp} \}^{\otimes 2}. \tag{3.10}$$

We take $\widehat{\mathbf{R}}_m$ as the first $k_r^{(m)}$ eigenvectors of $\mathbf{X}_{R,m}$, and $\widehat{\mathbf{C}}_m$ as the first $k_c^{(m)}$ eigenvectors of $\mathbf{X}_{C,m}$.

The local common factor $\mathbf{F}_t^{(m)}$ is estimated using least square method:

$$\widehat{\mathbf{F}}_t^{(m)} := (\widehat{\mathbf{R}}_m' \widehat{\mathbf{Q}}_m^{\perp} \widehat{\mathbf{Q}}_m^{\perp\prime} \widehat{\mathbf{R}}_m)^{-1} \widehat{\mathbf{R}}_m' \widehat{\mathbf{Q}}_m^{\perp} \widehat{\mathbf{Q}}_m^{\perp\prime} \breve{\mathbf{Y}}_t^{(m)} \widehat{\mathbf{C}}_m.$$

Define $\breve{\mathbf{Y}}_{G,t}^{(m)} := \breve{\mathbf{Y}}_t^{(m)} - \widehat{\mathbf{R}}_m \widehat{\mathbf{F}}_t^{(m)} \widehat{\mathbf{C}}_m'$, and

$$\widehat{\mathbf{Z}}_{G,m} := \left(\mathbf{vec}(\widehat{\mathbf{Q}}_m' \breve{\mathbf{Y}}_{G,1}^{(m)} \widehat{\mathbf{J}}_m), \ldots, \mathbf{vec}(\widehat{\mathbf{Q}}_m' \breve{\mathbf{Y}}_{G,T}^{(m)} \widehat{\mathbf{J}}_m) \right).$$

We first choose a reference group $m_0 \in [M]$. For each $m \neq m_0$, define

$$\widehat{\mathbf{\Xi}}_m := \widehat{\mathbf{Z}}_{G,m_0} \widehat{\mathbf{Z}}_{G,m}' (\widehat{\mathbf{Z}}_{G,m} \widehat{\mathbf{Z}}_{G,m}')^{-1}.$$

We obtain a two-sided alignment by solving $\min_{\mathbf{\Xi}_{Q,m} \in \mathbb{R}^{k_r \times k_r}, \mathbf{\Xi}_{J,m} \in \mathbb{R}^{k_c \times k_c}} \left\| \mathbf{\Xi}_{J,m} \otimes \mathbf{\Xi}_{Q,m} - \widehat{\mathbf{\Xi}}_m \right\|_F^2$ (see e.g., Section 6 of Loan, 2000). We estimate $\mathbf{\Xi}_{Q,m}$ and $\mathbf{\Xi}_{J,m}$ using $\mathbf{vec}(\widehat{\mathbf{\Xi}}_{Q,m}) = \widehat{\sigma}_{\Xi,m}^{1/2} \widehat{\mathbf{u}}_{\Xi,m}$ and $\mathbf{vec}(\widehat{\mathbf{\Xi}}_{J,m}) = \widehat{\sigma}_{\Xi,m}^{1/2} \widehat{\mathbf{v}}_{\Xi,m}$ respectively, where $\widehat{\sigma}_{\Xi,m}$, $\widehat{\mathbf{u}}_{\Xi,m}$ and $\widehat{\mathbf{v}}_{\Xi,m}$ are the largest singular value, the corresponding left and right singular vectors of $\mathcal{R}(\widehat{\mathbf{\Xi}}_m)$, respectively; the matrix $\mathcal{R}(\mathbf{W})$ for a block matrix $\mathbf{W} = (\mathbf{W}_{ij})_{i \in [k_c], j \in [k_c]}$, with each $\mathbf{W}_{ij}$ of size $k_r \times k_r$, is defined as

$$\mathcal{R}(\mathbf{W}) := (\mathbf{vec}(\mathbf{W}_{11}), \ldots, \mathbf{vec}(\mathbf{W}_{k_c 1}), \ldots, \mathbf{vec}(\mathbf{W}_{1 k_c}), \ldots, \mathbf{vec}(\mathbf{W}_{k_c k_c})).$$

For the reference group, set $\widehat{\mathbf{\Xi}}_{Q,m_0} := \mathbf{I}_{k_r}$ and $\widehat{\mathbf{\Xi}}_{J,m_0} := \mathbf{I}_{k_c}$. The global common factor series is then estimated as

$$\widehat{\mathbf{G}}_t := M^{-1} \sum_{m=1}^{M} \widehat{\mathbf{\Xi}}_{Q,m} \widehat{\mathbf{Q}}_m' \breve{\mathbf{Y}}_{G,t}^{(m)} \widehat{\mathbf{J}}_m \widehat{\mathbf{\Xi}}_{J,m}'.$$

We then re-estimate $\mathbf{J}_m \otimes \mathbf{Q}_m$ by least squares:

$$\widehat{\mathbf{J}_m \otimes \mathbf{Q}_m} := \big(\mathbf{vec}(\breve{\mathbf{Y}}_{G,1}^{(m)}), \ldots, \mathbf{vec}(\breve{\mathbf{Y}}_{G,T}^{(m)})\big)\big(\mathbf{vec}(\widehat{\mathbf{G}}_1), \ldots, \mathbf{vec}(\widehat{\mathbf{G}}_T)\big)'[(\mathbf{vec}(\widehat{\mathbf{G}}_1), \ldots, \mathbf{vec}(\widehat{\mathbf{G}}_T))^{\otimes 2}]^{-1}.$$

Finally, to solve $\min_{\mathbf{Q}_m, \mathbf{J}_m} \big\|\mathbf{J}_m \otimes \mathbf{Q}_m - \widehat{\mathbf{J}_m \otimes \mathbf{Q}_m}\big\|_F^2$, we estimate $\mathbf{Q}_m$ and $\mathbf{J}_m$ by setting $\mathbf{vec}(\widetilde{\mathbf{Q}}_m) = \widehat{\sigma}_{Q,m}^{1/2} \widehat{\mathbf{u}}_{Q,m}$ and $\mathbf{vec}(\widetilde{\mathbf{J}}_m) = \widehat{\sigma}_{Q,m}^{1/2} \widehat{\mathbf{v}}_{J,m}$ respectively, where $\widehat{\sigma}_{Q,m}$, $\widehat{\mathbf{u}}_{Q,m}$ and $\widehat{\mathbf{v}}_{J,m}$ are the largest singular value, and the corresponding left and right singular vectors of $\mathcal{R}(\widehat{\mathbf{J}_m \otimes \mathbf{Q}_m})$ respectively.

Different from the global time series estimators proposed in Zhang et al. (2025), our proposed $\widehat{\boldsymbol{\alpha}}_t$, $\widehat{\boldsymbol{\beta}}_t$ in Step 2 and $\widehat{\mathbf{G}}_t$ in Step 3 are all independent of the index m, with all associated loading matrices re-estimated based on these truly global time series estimators.

Remark 3 *We stress that the grand mean time series* $\{c_t^{(m)}\}$ *and the main effects time series* $\{\mathbf{a}_t^{(m)}\}$ *and* $\{\mathbf{b}_t^{(m)}\}$ *themselves can be nonstationary, which is an important feature for MEFM in Lam and Cen (2025). The global or local grand means and main effects can also be nonstationary, as long as Conditions (ME1) and (ME2) in Section 4.1 are satisfied.*

Remark 4 *The two-sided alignment step is needed because, for each* m, $\widehat{\mathbf{Q}}_m' \breve{\mathbf{Y}}_{G,t}^{(m)} \widehat{\mathbf{J}}_m$ *is expressed in group-specific row and column coordinates. In particular, if* $\widehat{\mathbf{Q}}_m' \breve{\mathbf{Y}}_{G,t}^{(m)} \widehat{\mathbf{J}}_m \approx \mathbf{L}_m \mathbf{G}_t \mathbf{S}_m'$, *then direct averaging yields* $M^{-1} \sum_{m=1}^{M} \mathbf{L}_m \mathbf{G}_t \mathbf{S}_m'$, *while* $M^{-1} \sum_{m=1}^{M} (\mathbf{S}_m \otimes \mathbf{L}_m)$ *need not be a Kronecker product. Hence, the average generally cannot be written as* $\mathbf{L}\mathbf{G}_t\mathbf{S}'$ *for a common pair* $(\mathbf{L}, \mathbf{S})$, *and the resulting coordinate change cannot in general be absorbed by the subsequent two-sided least-squares refitting.*

At the population level, $\mathbf{G}_t$ *is represented in the coordinate systems of group* m *and* m_0 *as* $\mathbf{L}_m \mathbf{G}_t \mathbf{S}_m'$ *and* $\mathbf{L}_{m_0} \mathbf{G}_t \mathbf{S}_{m_0}'$, *respectively. The vectorized coordinate change from group* m *to group* m_0 *is then* $(\mathbf{S}_{m_0}\mathbf{S}_m^{-1}) \otimes (\mathbf{L}_{m_0}\mathbf{L}_m^{-1})$, *since applying this transformation to* $\mathbf{vec}(\mathbf{L}_m \mathbf{G}_t \mathbf{S}_m')$ *yields* $\mathbf{vec}(\mathbf{L}_{m_0} \mathbf{G}_t \mathbf{S}_{m_0}')$. *Under the rearrangement operator,*

$$\mathcal{R}\left((\mathbf{S}_{m_0}\mathbf{S}_m^{-1}) \otimes (\mathbf{L}_{m_0}\mathbf{L}_m^{-1})\right) = \mathbf{vec}(\mathbf{L}_{m_0}\mathbf{L}_m^{-1})\mathbf{vec}(\mathbf{S}_{m_0}\mathbf{S}_m^{-1})'.$$

Hence, the leading left and right singular vectors of $\mathcal{R}(\widehat{\mathbf{\Xi}}_m)$ *estimate the row and column alignment matrices, bringing the representation from group* m *into the coordinate system of group* m_0 *before averaging.*

Remark 5 *The reference group in the global alignment can be chosen arbitrarily in principle. In numerical implementation, we choose the group with the largest projected-core norm, for instance the largest value of* $\sum_{t=1}^{T}\left\|\widehat{\mathbf{Q}}_m'\check{\mathbf{Y}}_{G,t}^{(m)}\widehat{\mathbf{J}}_m\right\|_F^2$ *to improve numerical stability and to avoid using a group whose global common component is nearly degenerate as the reference. A more formal procedure for detecting whether a group contains nonzero global common factors is given in Section 4.3.*

3.4 Rank estimation for all loading matrices

The rank of all loading matrices in model (3.1) are assumed known so far. We propose the following eigenvalue-ratio based estimators for r_1, $r_2^{(m)}$, ℓ_1, $\ell_2^{(m)}$, k_r, k_c, $k_r^{(m)}$ and $k_c^{(m)}$ respectively:

$$\widehat{r}_1 := \underset{j\in[p_m]}{\arg\min}\frac{\max_{m\in[M]}\lambda_{j+1}(\widehat{\mathbf{\Omega}}_{A_1,m})}{\max_{m\in[M]}\lambda_j(\widehat{\mathbf{\Omega}}_{A_1,m})},\quad \widehat{r}_2^{(m)} := \underset{j\in[p_m]}{\arg\min}\frac{\lambda_{j+1}(\widehat{\mathbf{\Omega}}_{A_2,m})}{\lambda_j(\widehat{\mathbf{\Omega}}_{A_2,m})},\quad \widehat{\ell}_1 := \underset{j\in[q_m]}{\arg\min}\frac{\max_{m\in[M]}\lambda_{j+1}(\widehat{\mathbf{\Omega}}_{B_1,m})}{\max_{m\in[M]}\lambda_j(\widehat{\mathbf{\Omega}}_{B_1,m})},$$
$$\widehat{\ell}_2^{(m)} := \underset{j\in[q_m]}{\arg\min}\frac{\lambda_{j+1}(\widehat{\mathbf{\Omega}}_{B_2,m})}{\lambda_j(\widehat{\mathbf{\Omega}}_{B_2,m})},\quad \widehat{k}_r := \underset{j\in[p_m]}{\arg\min}\frac{\max_{m\in[M]}\lambda_{j+1}(\mathbf{X}_{Q,m})}{\max_{m\in[M]}\lambda_j(\mathbf{X}_{Q,m})},\quad \widehat{k}_c := \underset{j\in[q_m]}{\arg\min}\frac{\max_{m\in[M]}\lambda_{j+1}(\mathbf{X}_{J,m})}{\max_{m\in[M]}\lambda_j(\mathbf{X}_{J,m})},$$
$$\widehat{k}_r^{(m)} := \underset{j\in[p_m]}{\arg\min}\frac{\lambda_{j+1}(\mathbf{X}_{R,m})}{\lambda_j(\mathbf{X}_{R,m})},\quad \widehat{k}_c^{(m)} := \underset{j\in[q_m]}{\arg\min}\frac{\lambda_{j+1}(\mathbf{X}_{C,m})}{\lambda_j(\mathbf{X}_{C,m})},$$

where $\widehat{\mathbf{\Omega}}_{A_1,m}$, $\widehat{\mathbf{\Omega}}_{B_1,m}$, $\widehat{\mathbf{\Omega}}_{A_2,m}$, $\widehat{\mathbf{\Omega}}_{B_2,m}$, $\mathbf{X}_{Q,m}$, $\mathbf{X}_{J,m}$, $\mathbf{X}_{R,m}$ and $\mathbf{X}_{C,m}$ are defined in (3.3), (3.4), (3.5), (3.8), (3.9) and (3.10), respectively.

4 Assumptions and Theoretical Results

4.1 Assumptions

We say $\{\mathbf{w}_t\}_{t\in[T]} \in \mathcal{G}_T(\{d_1,\ldots,d_m\}|\{\mathbf{x}_i, i\in\mathbb{Z}\})$ if the process $\{\mathbf{w}_t\}_{t\in[T]}$ is piecewise-stationary with m stationary segments, each with length of order T. In particular, for t within the j-th segment, $\mathbf{w}_t$ can be expressed as

$$\mathbf{w}_t = \sum_{w\geq 0}\psi_j(w)\mathbf{x}_{t-w},\quad \psi_j(0)=1,\quad \psi_j(w) := w^{d_j-1}l_j(w) \text{ for } w\geq 1,$$

where $d_j < 1/2$, and $l_j(\cdot):\mathbb{R}\mapsto\mathbb{R}$ is a function such that $l_j(w)$ converges to a constant as $w\to\infty$. All elements in $\{\mathbf{x}_w\}$ have mean 0 and are independent and identically distributed with bounded 4th order moments. To facilitate notations under the regime of $\mathcal{G}_T(\cdot)$, define

$$R(T,d) := \begin{cases} T^{-1}, & d<1/4,\\ T^{-1}\log T, & d=1/4,\\ T^{4d-2}, & 1/4<d<1/2. \end{cases} \tag{4.1}$$

(ME1) (Main effect (uniform) factor strength). We assume that for each $m \in [M]$, as $p_m, q_m \to \infty$,

$$\mathbf{A}_{1m}'\mathbf{A}_{1m}/p_m^{\gamma_r} \to \mathbf{\Sigma}_{A_1,m}, \qquad \mathbf{B}_{1m}'\mathbf{B}_{1m}/q_m^{\gamma_c} \to \mathbf{\Sigma}_{B_1,m},$$
$$\mathbf{A}_{2m}'\mathbf{A}_{2m}/p_m^{\gamma_r^{(m)}} \to \mathbf{\Sigma}_{A_2,m}, \qquad \mathbf{B}_{2m}'\mathbf{B}_{2m}/q_m^{\gamma_c^{(m)}} \to \mathbf{\Sigma}_{B_2,m}.$$

where $\gamma_r, \gamma_r^{(m)}, \gamma_c, \gamma_c^{(m)}$ are all in $(1/2, 1]$. The matrices $\mathbf{\Sigma}_{A_1,m}$, $\mathbf{\Sigma}_{B_1,m}$, $\mathbf{\Sigma}_{A_2,m}$, and $\mathbf{\Sigma}_{B_2,m}$ are positive definite with eigenvalues bounded away from zero and infinity for all $m \in [M]$.

(ME2) (Time series in main effects). The time series $\{\boldsymbol{\alpha}_t\}, \{\boldsymbol{\alpha}_t^{(m)}\}, \{\boldsymbol{\beta}_t\}$ and $\{\boldsymbol{\beta}_t^{(m)}\}$ for $t \in [T]$ are all in $\mathcal{G}_T$ with potentially different set and number of piecewise-stationary segments, different set of d_j's and l_j's, and defined with time series $\{\mathbf{x}_{\alpha,t}\}, \{\mathbf{x}_{\alpha,t}^{(m)}\}, \{\mathbf{x}_{\beta,t}\}$ and $\{\mathbf{x}_{\beta,t}^{(m)}\}$ respectively. Also, $\{\mathbf{x}_{\alpha,t}^{(m)}\}$ and $\{\mathbf{x}_{\beta,t}^{(m)}\}$ are independent of $\{\mathbf{x}_{\alpha,t}^{(n)}\}$ and $\{\mathbf{x}_{\beta,t}^{(n)}\}$ respectively for $m \neq n$.

(E1) (Decomposition of error). We assume that

$$\mathbf{E}_t^{(m)} = \mathbf{Q}_{e,m}\mathbf{G}_{e,t}\mathbf{J}_{e,m}' + \mathbf{R}_{e,m}\mathbf{F}_{e,t}^{(m)}\mathbf{C}_{e,m}' + \mathbf{\Sigma}_{\epsilon,m} \circ \boldsymbol{\epsilon}_t^{(m)},$$

where $\mathbf{G}_{e,t}$ and $\mathbf{F}_{e,t}^{(m)}$ are $k_{e,r} \times k_{e,c}$ and $k_{e,r}^{(m)} \times k_{e,c}^{(m)}$ matrices containing independent elements with mean 0 and variance 1, respectively. The matrix $\boldsymbol{\epsilon}_t^{(m)} \in \mathbb{R}^{p_m \times q_m}$ consists of independent entries with mean 0 and variance 1. The series $\{\mathbf{G}_{e,t}\}$, $\{\mathbf{F}_{e,t}^{(m)}\}$ and $\{\boldsymbol{\epsilon}_t^{(m)}\}$ are mutually independent. The matrix $\mathbf{\Sigma}_{\epsilon,m}$ contains standard deviations, uniformly bounded away from 0 and infinity, of the corresponding elements in $\boldsymbol{\epsilon}_t^{(m)}$. Moreover, the error loadings are all (approximately) sparse matrices uniformly across all groups, i.e., for all $m \in [M]$, $\|\mathbf{Q}_{e,m}\|_1 = \|\mathbf{J}_{e,m}\|_1 = \|\mathbf{R}_{e,m}\|_1 = \|\mathbf{C}_{e,m}\|_1 = O(1)$, with $k_{e,r}, k_{e,c}, k_{e,r}^{(m)}, k_{e,c}^{(m)} = O(1)$. Furthermore, assume $\mathbf{Q}_{e,m}\mathbf{Q}_{e,n}' = \mathbf{0}$ or $\mathbf{J}_{e,m}\mathbf{J}_{e,n}' = \mathbf{0}$ for each $m \neq n$.

(E2) (Time series in $\mathbf{E}_t^{(m)}$). There is $\mathbf{X}_{e,t}$ the same dimension as $\mathbf{G}_{e,t}$, $\mathbf{X}_{e,t}^{(m)}$ the same dimension as $\mathbf{F}_{e,t}^{(m)}$ and $\mathbf{X}_{\epsilon,t}^{(m)}$ the same dimension as $\boldsymbol{\epsilon}_t^{(m)}$ such that $\mathbf{G}_{e,t} = \sum_{w\geq 0} a_{e,w}\mathbf{X}_{e,t-w}$, $\mathbf{F}_{e,t}^{(m)} = \sum_{w\geq 0} a_{e,w}^{(m)}\mathbf{X}_{e,t-w}^{(m)}$ and $\boldsymbol{\epsilon}_t^{(m)} = \sum_{w\geq 0} a_{\epsilon,w}^{(m)}\mathbf{X}_{\epsilon,t-w}^{(m)}$, with $\{\mathbf{X}_{e,t}\}$, $\{\mathbf{X}_{e,t}^{(m_1)}\}$, $\{\mathbf{X}_{e,t}^{(m_2)}\}$, $\{\mathbf{X}_{\epsilon,t}^{(n_1)}\}$ and $\{\mathbf{X}_{\epsilon,t}^{(n_2)}\}$ independent of each other for all $m_1, m_2, n_1, n_2 \in [M]$. The series $\{\mathbf{X}_{e,t}\}$ and $\{\mathbf{X}_{e,t}^{(m)}\}$ have independent elements for each $m \in [M]$, while $\{\mathbf{X}_{\epsilon,t}^{(m)}\}$ has i.i.d. elements for each $m \in [M]$, all with mean 0 and variance 1 with uniformly bounded fourth order moments. The time series $\{\mathbf{X}_{e,t}\}$ $\{\mathbf{X}_{e,t}^{(m)}\}$, and $\{\mathbf{X}_{\epsilon,t}^{(m)}\}$ are independent of all main effect and factor series in model (3.1) for each $m \in [M]$.

The coefficients $a_{e,w}$, $a_{e,w}^{(m)}$ and $a_{\epsilon,w}^{(m)}$ are such that $\sum_{w\geq 0} a_{e,w}^2 = \sum_{w\geq 0} a_{e,w}^{(m)2} = \sum_{w\geq 0} a_{\epsilon,w}^{(m)2} = 1$ and $\sum_{w\geq 0} |a_{e,w}|, \sum_{w\geq 0} |a_{e,w}^{(m)}|, \sum_{w\geq 0} |a_{\epsilon,w}^{(m)}| \leq c$ for some constant $c > 0$.

(F1) (Time series in $\mathbf{G}_t$ and $\mathbf{F}_t^{(m)}$). For $t \in [T]$, define $\mathbf{z}_{G,t} := \mathbf{vec}(\mathbf{G}_t')$, and $\mathbf{z}_{F,t}^{(m)} := \mathbf{vec}(\mathbf{F}_t^{(m)'})$ for each

$m \in [M]$. The time series $\{\mathbf{z}_{G,t}\}$ and $\{\mathbf{z}_{F,t}^{(m)}\}$ are in $\mathcal{G}_T$ with potentially different sets and numbers of piecewise-stationary segments, different sets of d_j's and l_j's, and defined with independent time series $\{\mathbf{x}_{G,t}\}$ and $\{\mathbf{x}_{F,t}^{(m)}\}$ respectively. Also, $\{\mathbf{x}_{F,t}^{(m)}\}$ is independent of $\{\mathbf{x}_{F,t}^{(n)}\}$ for $m \neq n$. Both $\{\mathbf{G}_t\}$ and $\{\mathbf{F}_t^{(m)}\}$ are independent of $\{\mathbf{E}_t^{(n)}\}$ for $m, n \in [M]$.

(L1) (Factor loading matrices) For each $m \in [M]$, define $\mathbf{Z}_q^{(m)} := \mathrm{diag}(\mathbf{Q}_m'\mathbf{Q}_m)$, $\mathbf{Z}_j^{(m)} := \mathrm{diag}(\mathbf{J}_m'\mathbf{J}_m)$, $\mathbf{Z}_r^{(m)} := \mathrm{diag}(\mathbf{R}_m'\mathbf{R}_m)$ and $\mathbf{Z}_c^{(m)} := \mathrm{diag}(\mathbf{C}_m'\mathbf{C}_m)$. We assume that

$$(\mathbf{Z}_q^{(m)})^{-1/2}\mathbf{Q}_m'\mathbf{Q}_m(\mathbf{Z}_q^{(m)})^{-1/2} \to \mathbf{\Sigma}_{Q,m}, \quad (\mathbf{Z}_j^{(m)})^{-1/2}\mathbf{J}_m'\mathbf{J}_m(\mathbf{Z}_j^{(m)})^{-1/2} \to \mathbf{\Sigma}_{J,m},$$
$$(\mathbf{Z}_r^{(m)})^{-1/2}\mathbf{R}_m'\mathbf{R}_m(\mathbf{Z}_r^{(m)})^{-1/2} \to \mathbf{\Sigma}_{R,m}, \quad (\mathbf{Z}_c^{(m)})^{-1/2}\mathbf{C}_m'\mathbf{C}_m(\mathbf{Z}_c^{(m)})^{-1/2} \to \mathbf{\Sigma}_{C,m},$$

where $\mathbf{\Sigma}_{Q,m}$, $\mathbf{\Sigma}_{J,m}$, $\mathbf{\Sigma}_{R,m}$ and $\mathbf{\Sigma}_{C,m}$ have eigenvalues uniformly bounded away from 0 and infinity for all $m \in [M]$. Also, we assume

$$(\mathbf{Z}_q^{(m)})_{ii} \asymp p_m^{\delta_{q,i}^{(m)}} \text{ for } i \in [k_r], \qquad (\mathbf{Z}_j^{(m)})_{ii} \asymp q_m^{\delta_{j,i}^{(m)}} \text{ for } i \in [k_c],$$
$$(\mathbf{Z}_r^{(m)})_{ii} \asymp p_m^{\delta_{r,i}^{(m)}} \text{ for } i \in [k_r^{(m)}], \qquad (\mathbf{Z}_c^{(m)})_{ii} \asymp q_m^{\delta_{c,i}^{(m)}} \text{ for } i \in [k_c^{(m)}],$$

where $\delta_{q,i}^{(m)}, \delta_{j,i}^{(m)}, \delta_{r,i}^{(m)}$ and $\delta_{c,i}^{(m)}$ are all in $(1/2, 1]$, ordered non-increasingly.

(S1) (Non-degeneracy and separation conditions) As $T \to \infty$, the eigenvalues of $T^{-1}\sum_{t=1}^T \mathbb{E}(\boldsymbol{\alpha}_t^{\otimes 2})$, $T^{-1}\sum_{t=1}^T \mathbb{E}(\boldsymbol{\beta}_t^{\otimes 2})$, $T^{-1}\sum_{t=1}^T \mathbb{E}(\boldsymbol{\alpha}_t^{(m)\otimes 2})$, $T^{-1}\sum_{t=1}^T \mathbb{E}(\boldsymbol{\beta}_t^{(m)\otimes 2})$, $T^{-1}\sum_{t=1}^T \mathbb{E}(\mathbf{vec}(\mathbf{G}_t)^{\otimes 2})$, and $T^{-1}\sum_{t=1}^T \mathbb{E}(\mathbf{vec}(\mathbf{F}_t^{(m)})^{\otimes 2})$ are all bounded away from zero and infinity over $m \in [M]$. In addition, for sufficiently large p_m and q_m, the singular values of

$$\mathbf{A}_{1m}^{\perp'}\mathbf{A}_{2m}(\mathbf{A}_{2m}'\mathbf{A}_{2m})^{-1/2}, \qquad \mathbf{B}_{1m}^{\perp'}\mathbf{B}_{2m}(\mathbf{B}_{2m}'\mathbf{B}_{2m})^{-1/2},$$
$$\mathbf{Q}_m^{\perp'}\mathbf{R}_m(\mathbf{R}_m'\mathbf{R}_m)^{-1/2}, \qquad \mathbf{J}_m^{\perp'}\mathbf{C}_m(\mathbf{C}_m'\mathbf{C}_m)^{-1/2}.$$

are bounded away from zero and infinity for each $m \in [M]$.

The overall structure of Assumptions (ME1)–(L1) broadly follows Lam and Cen (2025), with modifications to accommodate the multilevel structure and the broader temporal class $\mathcal{G}_T$. Assumption (ME1) allows global and local main effects to differ in strength, but assumes equal strength within each loading matrix for simplicity. By contrast, (L1) allows heterogeneous strengths across loading directions in the common components. Assumptions (ME2) and (F1) place the main-effect and common-component factors in $\mathcal{G}_T$, allowing changes in dynamics as well as short- and long-memory dependence. Under (F1), the entries of $\{\mathbf{z}_{G,t}\}$ and $\{\mathbf{z}_{F,t}^{(m)}\}$ are independent, while the cross-sectional dependence is accommodated through

$\mathbf{\Sigma}_{Q,m}$, $\mathbf{\Sigma}_{J,m}$, $\mathbf{\Sigma}_{R,m}$ and $\mathbf{\Sigma}_{C,m}$ in (L1). Assumptions (E1)–(E2) allow cross-sectional and serial dependence in the errors, while the cross-group orthogonality condition in (E1) prevents the errors from generating spurious global signals. Similar to Conditions 2 and 3 in Zhang et al. (2025), Assumption (S1) ensures non-degenerate factor processes and sufficient separation between the local and global loading spaces, so that the local signals are preserved after projecting out the global loading spaces.

Remark 6 *The orthogonality condition* $\mathbf{Q}_{e,m}\mathbf{Q}'_{e,n} = \mathbf{0}$ *or* $\mathbf{J}_{e,m}\mathbf{J}'_{e,n} = \mathbf{0}$ *for each* $m \neq n$ *in (E1) is imposed to ensure that the error matrices are uncorrelated across groups, as required by (IC2). However, this condition could be relaxed, in which case the model would be asymptotically identified. Such a relaxation would introduce additional terms into the cross-group error products and lead to more complicated proofs and convergence rates. We do not pursue the corresponding theoretical extension here, but in simulation in subsection 6.1, this condition is not imposed.*

4.2 Rate of convergence for estimators

We establish the asymptotic rates of convergence for the estimators depicted in Section 3.3. First, we consider the estimators of grand means and main effects.

Theorem 2 (Consistency of grand mean and main effect estimators) *Let Assumptions (IC1), (IC2), (E1) and (E2) hold. Then, for each* $m \in [M]$ *and* $t \in [T]$,

$$(\widehat{c}_t^{(m)} - c_t^{(m)})^2 = O_P\Big(\frac{1}{p_m q_m}\Big), \quad \frac{1}{p_m}\big\|\widehat{\mathbf{a}}_t^{(m)} - \mathbf{a}_t^{(m)}\big\|^2 = O_P\Big(\frac{1}{q_m}\Big), \quad \frac{1}{q_m}\big\|\widehat{\mathbf{b}}_t^{(m)} - \mathbf{b}_t^{(m)}\big\|^2 = O_P\Big(\frac{1}{p_m}\Big).$$

Moreover, for each $m \in [M]$ *and* $t \in [T]$,

$$\big(\widehat{\mu}_t - \mu_t\big)^2 = O_P\Big(\max_{m\in[M]} \frac{1}{p_m q_m}\Big), \qquad \big(\widehat{\mu}_t^{(m)} - \mu_t^{(m)}\big)^2 = O_P\Big(\max_{m\in[M]} \frac{1}{p_m q_m}\Big).$$

Theorem 2 demonstrates the performance of direct moment estimation and reflects the gain from cross-sectional averaging.

(R1) (Rate conditions for main effects) For each $m \in [M]$, define $\nu_{r,m} := p_m^{\gamma_r^{(m)} - \gamma_r}$ and $\nu_{c,m} := q_m^{\gamma_c^{(m)} - \gamma_c}$. Let $d_{r,\star}$ be the largest memory parameter across all elements of $\boldsymbol{\alpha}_t$ and $\boldsymbol{\alpha}_t^{(m)}$, $m \in [M]$, with $d_{c,\star}$ defined analogously for $\boldsymbol{\beta}_t$ and $\boldsymbol{\beta}_t^{(m)}$. We assume that

$$\max_{m\in[M]} \left\{ \frac{p_m^{1-\gamma_r}}{q_m}, \frac{p_m^{1-\gamma_r^{(m)}}}{q_m} \right\} = o(1), \qquad R(T, d_{r,\star}) \max_{\substack{m,n\in[M]\\ m\neq n}} \Big\{ (\nu_{r,m} \vee \frac{1}{\nu_{r,m}})(1 \vee \nu_{r,n}) \Big\} = o(1),$$

$$\max_{m\in[M]} \left\{ \frac{q_m^{1-\gamma_c}}{p_m}, \frac{q_m^{1-\gamma_c^{(m)}}}{p_m} \right\} = o(1), \qquad R(T, d_{c,\star}) \max_{\substack{m,n\in[M]\\ m\neq n}} \Big\{ (\nu_{c,m} \vee \frac{1}{\nu_{c,m}})(1 \vee \nu_{c,n}) \Big\} = o(1).$$

The quantities $\nu_{r,m}$ and $\nu_{c,m}$ measure the local main effect strength relative to the corresponding global effect. Assumption (R1) controls how this relative strength can vary with the matrix dimensions and temporal dependence. In particular, (R1) automatically holds when all main effect factors are pervasive.

Theorem 3 (Consistency of initial global main effect loading estimators) *Let the assumptions of Theorem 2 hold. Further assume Assumptions (ME1), (ME2), (R1) and (S1). For each $m \in [M]$, define $\breve{\mathbf{A}}_{1m} := \mathbf{A}_{1m}(\mathbf{A}_{1m}'\mathbf{A}_{1m})^{-1/2}$ and $\breve{\mathbf{B}}_{1m} := \mathbf{B}_{1m}(\mathbf{B}_{1m}'\mathbf{B}_{1m})^{-1/2}$. Then there exist asymptotically orthogonal matrices $\mathbf{H}_{A_1,m}$ and $\mathbf{H}_{B_1,m}$ with $\mathbf{H}_{A_1,m}\mathbf{H}_{A_1,m}' = \mathbf{I} + o_P(1)$ and $\mathbf{H}_{B_1,m}\mathbf{H}_{B_1,m}' = \mathbf{I} + o_P(1)$, such that*

$$\big\|\widehat{\mathbf{A}}_{1m} - \breve{\mathbf{A}}_{1m}\mathbf{H}_{A_1,m}\big\|_F^2 = O_P(\eta_{A_{1m}}),\ \eta_{A_{1m}} := R(T,d_{r,\star})\nu_{r,m}\max_{n\in[M]\setminus\{m\}}\nu_{r,n} + \frac{p_m^{1-\gamma_r}}{Tq_m}\big(1 \vee \max_{n\in[M]\setminus\{m\}}\nu_{r,n}\big),$$
$$\big\|\widehat{\mathbf{B}}_{1m} - \breve{\mathbf{B}}_{1m}\mathbf{H}_{B_1,m}\big\|_F^2 = O_P(\eta_{B_{1m}}),\ \eta_{B_{1m}} := R(T,d_{c,\star})\nu_{c,m}\max_{n\in[M]\setminus\{m\}}\nu_{c,n} + \frac{q_m^{1-\gamma_c}}{Tp_m}\big(1 \vee \max_{n\in[M]\setminus\{m\}}\nu_{c,n}\big).$$

Theorem 4 (Consistency of local main effect loading and factor estimators) *Let the assumptions of Theorem 3 hold. For each $m \in [M]$, define $\breve{\mathbf{A}}_{2m} := \mathbf{A}_{2m}(\mathbf{A}_{2m}'\mathbf{A}_{2m})^{-1/2}$ and $\breve{\mathbf{B}}_{2m} := \mathbf{B}_{2m}(\mathbf{B}_{2m}'\mathbf{B}_{2m})^{-1/2}$. Then there exist asymptotically orthogonal matrices $\mathbf{H}_{A_2,m}$ and $\mathbf{H}_{B_2,m}$ such that*

$$\big\|\widehat{\mathbf{A}}_{2m} - \breve{\mathbf{A}}_{2m}\mathbf{H}_{A_2,m}\big\|_F^2 = O_P\Big(R(T,d_{r,\star})\frac{\max_{n\in[M]\setminus\{m\}}\nu_{r,n}}{\nu_{r,m}} + \frac{p_m^{1-\gamma_r}}{Tq_m}\frac{\big(1\vee\max_{n\in[M]}\nu_{r,n}\big)}{\nu_{r,m}^2} + \frac{p_m^{1-2\gamma_r^{(m)}}}{q_m^2}\Big),$$
$$\big\|\widehat{\mathbf{B}}_{2m} - \breve{\mathbf{B}}_{2m}\mathbf{H}_{B_2,m}\big\|_F^2 = O_P\Big(R(T,d_{c,\star})\frac{\max_{n\in[M]\setminus\{m\}}\nu_{c,n}}{\nu_{c,m}} + \frac{q_m^{1-\gamma_c}}{Tp_m}\frac{\big(1\vee\max_{n\in[M]}\nu_{c,n}\big)}{\nu_{c,m}^2} + \frac{q_m^{1-2\gamma_c^{(m)}}}{p_m^2}\Big).$$

Moreover, for each $m \in [M]$ and $t \in [T]$, define $\breve{\boldsymbol{\alpha}}_t^{(m)} := (\mathbf{A}_{2m}'\mathbf{A}_{2m})^{1/2}\boldsymbol{\alpha}_t^{(m)}$ and $\breve{\boldsymbol{\beta}}_t^{(m)} := (\mathbf{B}_{2m}'\mathbf{B}_{2m})^{1/2}\boldsymbol{\beta}_t^{(m)}$. Then,

$$\frac{1}{p_m^{\gamma_r^{(m)}}}\big\|\widehat{\boldsymbol{\alpha}}_t^{(m)} - \mathbf{H}_{A_2,m}^{-1}\breve{\boldsymbol{\alpha}}_t^{(m)}\big\|^2 = O_P\Big(\frac{1}{p_m^{\gamma_r^{(m)}}q_m} + R(T,d_{r,\star})\frac{\max_{n\in[M]\setminus\{m\}}\nu_{r,n}}{1\wedge\nu_{r,m}} + \frac{p_m^{1-\gamma_r^{(m)}}}{Tq_m}\frac{1\vee\max_{n\in[M]}\nu_{r,n}}{1\wedge\nu_{r,m}}\Big),$$
$$\frac{1}{q_m^{\gamma_c^{(m)}}}\big\|\widehat{\boldsymbol{\beta}}_t^{(m)} - \mathbf{H}_{B_2,m}^{-1}\breve{\boldsymbol{\beta}}_t^{(m)}\big\|^2 = O_P\Big(\frac{1}{q_m^{\gamma_c^{(m)}}p_m} + R(T,d_{c,\star})\frac{\max_{n\in[M]\setminus\{m\}}\nu_{c,n}}{1\wedge\nu_{c,m}} + \frac{q_m^{1-\gamma_c^{(m)}}}{Tp_m}\frac{1\vee\max_{n\in[M]}\nu_{c,n}}{1\wedge\nu_{c,m}}\Big).$$

Theorem 5 (Consistency of global main effect factor and refitted loading estimators) *Let the assumptions in Theorem 4 hold. Define $\breve{\boldsymbol{\alpha}}_t := (T^{-1}\sum_{s=1}^T\boldsymbol{\alpha}_s\boldsymbol{\alpha}_s')^{-1/2}\boldsymbol{\alpha}_t$ and $\breve{\boldsymbol{\beta}}_t := (T^{-1}\sum_{s=1}^T\boldsymbol{\beta}_s\boldsymbol{\beta}_s')^{-1/2}\boldsymbol{\beta}_t$. Then there exist asymptotically orthogonal matrices $\mathbf{H}_\alpha$ and $\mathbf{H}_\beta$ such that, for each $t \in [T]$,*

$$\big\|\widehat{\boldsymbol{\alpha}}_t - \mathbf{H}_\alpha^{-1}\breve{\boldsymbol{\alpha}}_t\big\|^2 = O_P\Big(\max_{m\in[M]}\Big\{\frac{1}{p_m^{\gamma_r}q_m} + R(T,d_{r,\star})(1\vee\nu_{r,m})\max_{n\in[M]\setminus\{m\}}\nu_{r,n} + \frac{p_m^{1-\gamma_r}}{Tq_m}\frac{1\vee\max_{n\in[M]}\nu_{r,n}}{1\wedge\nu_{r,m}}\Big\}\Big),$$
$$\big\|\widehat{\boldsymbol{\beta}}_t - \mathbf{H}_\beta^{-1}\breve{\boldsymbol{\beta}}_t\big\|^2 = O_P\Big(\max_{m\in[M]}\Big\{\frac{1}{q_m^{\gamma_c}p_m} + R(T,d_{c,\star})(1\vee\nu_{c,m})\max_{n\in[M]\setminus\{m\}}\nu_{c,n} + \frac{q_m^{1-\gamma_c}}{Tp_m}\frac{1\vee\max_{n\in[M]}\nu_{c,n}}{1\wedge\nu_{c,m}}\Big\}\Big).$$

Moreover, let $\eta_{A_{1m}}$ and $\eta_{B_{1m}}$ denote the rates in Theorem 3, and η_α and η_β denote the rates in the two

factor bounds above. Define $\breve{\breve{\mathbf{A}}}_{1m} := \mathbf{A}_{1m}(T^{-1}\sum_{t=1}^{T}\boldsymbol{\alpha}_t\boldsymbol{\alpha}_t')^{1/2}$ *and* $\breve{\breve{\mathbf{B}}}_{1m} := \mathbf{B}_{1m}(T^{-1}\sum_{t=1}^{T}\boldsymbol{\beta}_t\boldsymbol{\beta}_t')^{1/2}$*. Then,*

$$\frac{1}{p_m^{\gamma_r}}\big\|\widetilde{\mathbf{A}}_{1m}-\breve{\breve{\mathbf{A}}}_{1m}\mathbf{H}_\alpha\big\|_F^2 = O_P\Big(\eta_{A_1,m}+\Big\{\frac{p_m^{1-\gamma_r}}{q_m}+R(T,d_{r,\star})(1\vee\nu_{r,m})\max_{n\in[M]\setminus\{m\}}\nu_{r,n}\Big\}\eta_\alpha\Big),$$
$$\frac{1}{q_m^{\gamma_c}}\big\|\widetilde{\mathbf{B}}_{1m}-\breve{\breve{\mathbf{B}}}_{1m}\mathbf{H}_\beta\big\|_F^2 = O_P\Big(\eta_{B_1,m}+\Big\{\frac{q_m^{1-\gamma_c}}{p_m}+R(T,d_{c,\star})(1\vee\nu_{c,m})\max_{n\in[M]\setminus\{m\}}\nu_{c,n}\Big\}\eta_\beta\Big).$$

Theorems 3–5 reveal how the relative strengths $\nu_{r,m}$ and $\nu_{c,m}$ induce a trade-off between global and local estimation. The refitted loading $\widetilde{\mathbf{A}}_{1m}$ inherits both the error of $\widehat{\mathbf{A}}_{1m}$ and that of the estimated global main effect factor series. Although refitting does not necessarily improve the loading convergence rate, it aligns the group-specific global loadings to a common temporal coordinate. This alignment substantially improves the estimation of the global main effect products, as demonstrated in subsection 6.1.2.

A useful benchmark is obtained when all main effect factors are pervasive and their memory parameters are below 1/4, as summarized in the following corollary.

Corollary 6 (Simplified rates of main effect estimators) *Let Assumptions (IC1), (IC2), (E1), (E2) and (S1) hold. Suppose that (ME1) and (ME2) hold with pervasive main effect factors, i.e.,* $\gamma_r = \gamma_c = \gamma_r^{(m)} = \gamma_c^{(m)} = 1$ *for all* $m \in [M]$*, and that all main effect memory parameters are less than* 1/4*. Then, for each* $m \in [M]$ *and* $t \in [T]$*,*

- $\big\|\widehat{\mathbf{A}}_{1m}-\check{\mathbf{A}}_{1m}\mathbf{H}_{A_1,m}\big\|_F^2 = O_P\big(1/T\big)$, $\big\|\widehat{\mathbf{B}}_{1m}-\check{\mathbf{B}}_{1m}\mathbf{H}_{B_1,m}\big\|_F^2 = O_P\big(1/T\big)$.
- $\big\|\widehat{\mathbf{A}}_{2m}-\check{\mathbf{A}}_{2m}\mathbf{H}_{A_2,m}\big\|_F^2 = O_P\big(1/T+1/(p_m q_m^2)\big)$, $\big\|\widehat{\mathbf{B}}_{2m}-\check{\mathbf{B}}_{2m}\mathbf{H}_{B_2,m}\big\|_F^2 = O_P\big(1/T+1/(q_m p_m^2)\big)$.
- $1/p_m\big\|\widehat{\boldsymbol{\alpha}}_t^{(m)}-\mathbf{H}_{A_2,m}^{-1}\check{\boldsymbol{\alpha}}_t^{(m)}\big\|^2 = O_P\big(1/T+1/(p_m q_m)\big)$, $1/q_m\big\|\widehat{\boldsymbol{\beta}}_t^{(m)}-\mathbf{H}_{B_2,m}^{-1}\check{\boldsymbol{\beta}}_t^{(m)}\big\|^2 = O_P\big(1/T+1/(p_m q_m)\big)$.
- $\big\|\widehat{\boldsymbol{\alpha}}_t-\mathbf{H}_\alpha^{-1}\check{\boldsymbol{\alpha}}_t\big\|^2 = O_P\big(1/T+\max_{n\in[M]}1/(p_n q_n)\big)$, $\big\|\widehat{\boldsymbol{\beta}}_t-\mathbf{H}_\beta^{-1}\check{\boldsymbol{\beta}}_t\big\|^2 = O_P\big(1/T+\max_{n\in[M]}1/(p_n q_n)\big)$.
- $1/p_m\big\|\widetilde{\mathbf{A}}_{1m}-\breve{\breve{\mathbf{A}}}_{1m}\mathbf{H}_\alpha\big\|_F^2 = O_P\big(1/T+1/q_m\max_{n\in[M]}1/(p_n q_n)\big)$, $1/q_m\big\|\widetilde{\mathbf{B}}_{1m}-\breve{\breve{\mathbf{B}}}_{1m}\mathbf{H}_\beta\big\|_F^2 = O_P\big(1/T+1/p_m\max_{n\in[M]}1/(p_n q_n)\big)$.

Before imposing some regularity conditions for Step 3 in Section 3.3, recall $\mathbf{v}_a^{(m)}$ and $\mathbf{v}_b^{(m)}$ therein. Then define $s_{Q,m} := \big\|\mathbf{Q}_m'\mathbf{v}_a^{(m)}\big\|$ and $s_{J,m} := \big\|\mathbf{J}_m'\mathbf{v}_b^{(m)}\big\|$, which are similarly seen in Chen and Lam (2024). In particular, we would have $s_{Q,m} \asymp p_m^{\delta_{q,1}/2}$ when $\mathbf{v}_a^{(m)}$ aligns with the column of $\mathbf{Q}_m$ corresponding to the strongest factor. In the following, we present some assumptions on the projection vectors $\mathbf{v}_a^{(m)}$ and $\mathbf{v}_b^{(m)}$.

(L2) (Cross-group alignment of projection directions) There exists an absolute constant $c > 0$ such that for all $m, n \in [M]$, with probability tending to one,

$$\big|\mathbf{v}_b^{(m)'}\mathbf{J}_m\mathbf{J}_n'\mathbf{v}_b^{(n)}\big| \ge c\,\|\mathbf{J}_m'\mathbf{v}_b^{(m)}\|\,\|\mathbf{J}_n'\mathbf{v}_b^{(n)}\|, \qquad \big|\mathbf{v}_a^{(m)'}\mathbf{Q}_m\mathbf{Q}_n'\mathbf{v}_a^{(n)}\big| \ge c\,\|\mathbf{Q}_m'\mathbf{v}_a^{(m)}\|\,\|\mathbf{Q}_n'\mathbf{v}_a^{(n)}\|.$$

(R2) (Signal dominance for global common component) Let $d_{\mathrm{fac},\star}$ be the largest long-memory exponent

among all elements of $\mathbf{G}_t$ and $\mathbf{F}_t^{(m)}$ for $m \in [M]$. Then for all $m, n \in [M]$ with $m \neq n$, assume that

$$p_m^{\delta_{r,1}^{(m)}-\delta_{q,1}^{(m)}} q_m^{\delta_{c,1}^{(m)}} = O(s_{J,m}^2), \quad p_n^{\delta_{r,1}^{(n)}-\delta_{q,1}^{(n)}} q_n^{\delta_{c,1}^{(n)}} = O(s_{J,n}^2), \quad p_m^{1-\delta_{q,1}^{(m)}} = O(s_{J,m}^2), \quad p_n^{1-\delta_{q,1}^{(n)}} = O(s_{J,n}^2),$$
$$q_m^{\delta_{c,1}^{(m)}-\delta_{j,1}^{(m)}} p_m^{\delta_{r,1}^{(m)}} = O(s_{Q,m}^2), \quad q_n^{\delta_{c,1}^{(n)}-\delta_{j,1}^{(n)}} p_n^{\delta_{r,1}^{(n)}} = O(s_{Q,n}^2), \quad q_m^{1-\delta_{j,1}^{(m)}} = O(s_{Q,m}^2), \quad q_n^{1-\delta_{j,1}^{(n)}} = O(s_{Q,n}^2).$$

Assumption (L2) formalizes that the selected projection directions align with the same latent global column and row factor directions across different groups. This assumption would hold if $\mathbf{v}_b^{(m)}$ and $\mathbf{v}_b^{(n)}$ are the leading singular directions of $\mathbf{J}_m$ and $\mathbf{J}_n$ correspond to the same strongest global factor. Our numerical results in Section 6.1.3 show that the iterative projection procedure in Section A estimate those vectors appropriately.

Theorem 7 (Consistency of local loading and factor estimators) *Let Assumptions (IC1), (E1), (E2), (F1), (L1), (L2), and (R2) hold. Define* $\breve{\mathbf{R}}_m := \mathbf{R}_m(\mathbf{R}_m'\mathbf{R}_m)^{-1/2}$, $\breve{\mathbf{C}}_m := \mathbf{C}_m(\mathbf{C}_m'\mathbf{C}_m)^{-1/2}$, *and*

$$\eta_{Q,m} := R(T, d_{fac,\star})\, p_m^{2(\delta_{q,1}^{(m)}-\delta_{q,k_r}^{(m)})} \left(\frac{\max_{n\in[M]\setminus\{m\}} s_{J,n}^2 p_n^{\delta_{q,1}^{(n)}}}{\max_{n\in[M]\setminus\{m\}} s_{J,n}^2 p_n^{\delta_{q,k_r}^{(n)}}} \right)^2,$$
$$\eta_{J,m} := R(T, d_{fac,\star})\, q_m^{2(\delta_{j,1}^{(m)}-\delta_{j,k_c}^{(m)})} \left(\frac{\max_{n\in[M]\setminus\{m\}} s_{Q,n}^2 q_n^{\delta_{j,1}^{(n)}}}{\max_{n\in[M]\setminus\{m\}} s_{Q,n}^2 q_n^{\delta_{j,k_c}^{(n)}}} \right)^2.$$

Further assume $\eta_{Q,m}, \eta_{J,m} = o(1)$. *Then there exist asymptotically orthogonal matrices* $\mathbf{H}_{R,m}$ *and* $\mathbf{H}_{C,m}$ *with* $\mathbf{H}_{R,m}\mathbf{H}_{R,m}' = \mathbf{I} + o_P(1)$ *and* $\mathbf{H}_{C,m}\mathbf{H}_{C,m}' = \mathbf{I} + o_P(1)$, *such that*

$$\left\|\widehat{\mathbf{R}}_m - \breve{\mathbf{R}}_m\mathbf{H}_{R,m}\right\|_F^2 = O_P(\Delta_{R,m}), \qquad \left\|\widehat{\mathbf{C}}_m - \breve{\mathbf{C}}_m\mathbf{H}_{C,m}\right\|_F^2 = O_P(\Delta_{C,m}), \quad \textit{where}$$
$$\Delta_{R,m} := p_m^{-2\delta_{r,k_r}^{(m)}} q_m^{-2\delta_{c,1}^{(m)}} \left\{ \left(p_m^{2\delta_{r,1}^{(m)}} q_m^{2\delta_{c,1}^{(m)}} + p_m^{1+\delta_{q,1}^{(m)}} q_m^{1+\delta_{j,1}^{(m)}}\right)\eta_{J,m} + \frac{p_m^{1+\delta_{r,1}^{(m)}} q_m^{1+\delta_{c,1}^{(m)}}}{T} + p_m^{2(1-\delta_{r,1}^{(m)})} + \frac{p_m^2 q_m}{T} \right\},$$
$$\Delta_{C,m} := q_m^{-2\delta_{c,k_c}^{(m)}} p_m^{-2\delta_{r,1}^{(m)}} \left\{ \left(q_m^{2\delta_{c,1}^{(m)}} p_m^{2\delta_{r,1}^{(m)}} + q_m^{1+\delta_{j,1}^{(m)}} p_m^{1+\delta_{q,1}^{(m)}}\right)\eta_{Q,m} + \frac{q_m^{1+\delta_{c,1}^{(m)}} p_m^{1+\delta_{r,1}^{(m)}}}{T} + q_m^{2(1-\delta_{c,1}^{(m)})} + \frac{q_m^2 p_m}{T} \right\}.$$

Further let Assumption (S1) hold, and define $\breve{\mathbf{F}}_t^{(m)} := \left(\mathbf{R}_m'\mathbf{R}_m\right)^{1/2}\mathbf{F}_t^{(m)}\left(\mathbf{C}_m'\mathbf{C}_m\right)^{1/2}$. *Then*

$$\frac{1}{p_m^{\delta_{r,1}^{(m)}} q_m^{\delta_{c,1}^{(m)}}} \left\|\widehat{\mathbf{F}}_t^{(m)} - \mathbf{H}_{R,m}^{-1}\breve{\mathbf{F}}_t^{(m)}\mathbf{H}_{C,m}\right\|_F^2 = O_P(\Delta_{R,m} + \Delta_{C,m} + \Delta_{F,m}), \quad \textit{where}$$
$$\Delta_{F,m} := \min\left(q_m^{1-2\delta_{c,1}^{(m)}} p_m^{1-\delta_{r,1}^{(m)}} \Delta_{R,m} + q_m^{1-2\delta_{c,1}^{(m)}} p_m^{-\delta_{r,1}^{(m)}}, p_m^{1-2\delta_{r,1}^{(m)}} q_m^{1-\delta_{c,1}^{(m)}} \Delta_{C,m} + p_m^{1-2\delta_{r,1}^{(m)}} q_m^{-\delta_{c,1}^{(m)}} \right).$$

Theorem 7 establishes the consistency of the local factor loading and factor estimators. The terms $\eta_{Q,m}$ and $\eta_{J,m}$ measure the error inherited from the estimated global loading directions, and the condition

$\eta_{Q,m}, \eta_{J,m} = o(1)$ ensures that the projected cross-group moment matrices $\mathbf{X}_{C,m}$ and $\mathbf{X}_{R,m}$ recover the local factor spaces consistently. When all common factors are strong, i.e., all δ's equal one, and dimensions are comparable, then $\Delta_{R,m}$ and $\Delta_{C,m}$ become $O_P(T^{-1})$. The additional Assumption (S1) then yields the rate for the local factor series $\widehat{\mathbf{F}}_t^{(m)}$, whose leading term combines $\Delta_{R,m}$, $\Delta_{C,m}$, and an additional term $\Delta_{F,m}$ accounting for the asymmetry between the row and column sides in the refitting.

Theorem 8 (Consistency of global loading and factor estimators) *Let all assumptions in Theorem 7 hold. Let $m_0 \in [M]$ be a fixed reference group, and define $\breve{\mathbf{G}}_t := \big(\mathbf{Q}_{m_0}'\mathbf{Q}_{m_0}\big)^{1/2}\mathbf{G}_t\big(\mathbf{J}_{m_0}'\mathbf{J}_{m_0}\big)^{1/2}$. For any $m \in [M]$, define*

$$\Delta_{G,m} := \min(p_m, q_m) + p_m^{\delta_{r,1}^{(m)}} q_m^{\delta_{c,1}^{(m)}}\big(\Delta_{R,m} + \Delta_{C,m} + \Delta_{F,m}\big) + \big(\eta_{Q,m} + \eta_{J,m}\big)p_m q_m,$$
$$S_m := \lambda_{\min}\big(\mathbf{Q}_m'\mathbf{Q}_m\big)\lambda_{\min}\big(\mathbf{J}_m'\mathbf{J}_m\big) \asymp p_m^{\delta_{q,k_r}^{(m)}} q_m^{\delta_{j,k_c}^{(m)}}.$$

and assume $\Delta_{G,m} = o\big\{\lambda_{\min}\big(\mathbf{Q}_m'\mathbf{Q}_m\big)\lambda_{\min}\big(\mathbf{J}_m'\mathbf{J}_m\big)\big\}$. Further assume that the reference group m_0 is chosen such that for any $m \in [M]$, $\lambda_{\min}\big(\mathbf{Q}_{m_0}'\mathbf{Q}_{m_0}\big)\lambda_{\min}\big(\mathbf{J}_{m_0}'\mathbf{J}_{m_0}\big) = O\big\{\lambda_{\min}\big(\mathbf{Q}_m'\mathbf{Q}_m\big)\lambda_{\min}\big(\mathbf{J}_m'\mathbf{J}_m\big)\big\}$. Then there exist asymptotically orthogonal matrices $\mathbf{H}_{Q,m_0}$ and $\mathbf{H}_{J,m_0}$ such that

$$\big\|\widehat{\mathbf{G}}_t - \mathbf{H}_{Q,m_0}'\breve{\mathbf{G}}_t\mathbf{H}_{J,m_0}\big\|_F^2 = O_P(\Delta_G), \quad \textit{where} \quad \Delta_G := \max_{m\in[M]}\Delta_{G,m} + \Big(\max_{m'\in[M]}\frac{\Delta_{G,m'}}{S_{m'}}\Big)\max_{m\in[M]}S_m.$$

Also, we have

$$\big\|\widetilde{\mathbf{Q}}_m - \mathbf{Q}_m\big(\mathbf{Q}_{m_0}'\mathbf{Q}_{m_0}\big)^{-1/2}\mathbf{H}_{Q,m_0}\big\|_F^2 = O_P(\Delta_{K,m}), \quad \textit{where} \quad \Delta_{K,m} := \Delta_G\frac{p_m^{\delta_{q,1}^{(m)}} q_m^{\delta_{j,1}^{(m)}}}{p_{m_0}^{2\delta_{q,k_r}^{(m_0)}} q_{m_0}^{2\delta_{j,k_c}^{(m_0)}}} + \frac{p_m q_m}{TS_{m_0}},$$

and similarly $\big\|\widetilde{\mathbf{J}}_m - \mathbf{J}_m\big(\mathbf{J}_{m_0}'\mathbf{J}_{m_0}\big)^{-1/2}\mathbf{H}_{J,m_0}\big\|_F^2 = O_P(\Delta_{K,m})$.

Built upon Theorem 7, Theorem 8 addresses the global common component. The quantity Δ_G has two parts. The first, $\max_{m\in[M]}\Delta_{G,m}$, aggregates the local-component estimation errors across groups. The second term in Δ_G arises from the two-sided alignment and the refitting step, where S_m measures the smallest squared signal in the m-th group's projected global loading matrices. The signal-dominance condition $\Delta_{G,m} = o(S_m)$ ensures that the perturbed Gram matrices in the alignment step remain well-conditioned, and the reference-group condition guarantees the alignment from any group to the reference group does not blow up. Under these conditions, the first part of the theorem shows that $\widehat{\mathbf{G}}_t$ is consistent for the aligned version $\mathbf{H}_{Q,m_0}'\breve{\mathbf{G}}_t\mathbf{H}_{J,m_0}$ of the global factor, which is infeasible by merely using the group-wise "global" part in Zhang et al. (2025). Then the theorem gives the rates for the refitted loadings $\widetilde{\mathbf{Q}}_m$ and $\widetilde{\mathbf{J}}_m$. In the strong-factor setting where all δ's equal one and all groups have comparable dimensions,

both terms reduce to $O_P(T^{-1})$, delivering consistency of the refitted global loadings up to the asymptotic orthogonal matrices $\mathbf{H}_{Q,m_0}$ and $\mathbf{H}_{J,m_0}$.

4.3 Detecting the existence of global common factors

So far we have assumed that the global factors in $\mathbf{G}_t$ drive the dynamics of each data set $\mathbf{Y}_t^{(m)}$ for $m \in [M]$. However, if we are exploring new data sets which may not share common dynamics with others, blindly applying Algorithm A.1 in Section A will give us a false impression that there are global factors driving the dynamics of these new data sets, when there may be none. It is important to first investigate if there are global factors driving the dynamics of any data set at all.

To this end, suppose we have $\widehat{\mathbf{Q}}_m$ and $\widehat{\mathbf{J}}_m$ as the output of Algorithm A.1 in Section A. Also, for $m \in [M]$, let $\mathbf{U}_m = (\mathbf{Q}_m, \mathbf{R}_m)$ and $\mathbf{V}_m = (\mathbf{J}_m, \mathbf{C}_m)$, and $\widehat{\mathbf{U}}_{m,\cdot c_m}$ and $\widehat{\mathbf{V}}_{m,\cdot d_m}$ be the c_mth and d_mth eigenvectors of

$$T^{-1}\sum_{t=1}^{T}\mathbf{M}_{p_m}\mathbf{Y}_t^{(m)}\mathbf{M}_{q_m}\mathbf{Y}_t^{(m)'}\mathbf{M}_{p_m} \quad \text{and} \quad T^{-1}\sum_{t=1}^{T}\mathbf{M}_{q_m}\mathbf{Y}_t^{(m)'}\mathbf{M}_{p_m}\mathbf{Y}_t^{(m)}\mathbf{M}_{q_m},$$

respectively (Lam and Cen, 2025). This means that we are using the data $\{\mathbf{Y}_t^{(m)}\}_{t\in[T]}$ only to estimate the MMEFM in (3.1) without distinguishing global and local factors.

Theorem 9 *Let all the conditions in Theorem 7 hold. Define the global active set as*

$$\mathcal{A}_G := \left\{m \in [M] : \textit{there exists } n \neq m \in [M] \textit{ such that } \{\mathbf{Q}_m\mathbf{G}_t\mathbf{J}_m'\}_{t\in[T]} \textit{ and } \{\mathbf{Q}_n\mathbf{G}_t\mathbf{J}_n'\}_{t\in[T]} \textit{ are nonzero}\right\}, \tag{4.2}$$

and the statistics

$$S_{G,m} := \frac{\sum_{n=1;n\neq m}^{M}\max_{c_m,c_n\in[k_r],d_m,d_n\in[k_c]}\left\{T^{-1}\sum_{t=1}^{T}\widehat{\mathbf{Q}}_{m,\cdot c_m}'\check{\mathbf{Y}}_t^{(m)}\widehat{\mathbf{J}}_{m,\cdot d_m}\widehat{\mathbf{J}}_{n,\cdot d_n}'\check{\mathbf{Y}}_t^{(n)'}\widehat{\mathbf{Q}}_{n,\cdot c_n}\right\}^2}{\max_{c_m\in[k_r+k_r^{(m)}],d_m\in[k_c+k_c^{(m)}]}\left(T^{-1}\sum_{t=1}^{T}\widehat{\mathbf{U}}_{m,\cdot c_m}'\check{\mathbf{Y}}_t^{(m)}\widehat{\mathbf{V}}_{m,\cdot d_m}\widehat{\mathbf{V}}_{m,\cdot d_m}'\check{\mathbf{Y}}_t^{(m)'}\widehat{\mathbf{U}}_{m,\cdot c_m}\right)^2}, \quad m \in [M]. \tag{4.3}$$

Then for any $m \in \mathcal{A}_G$ and $k \notin \mathcal{A}_G$, assuming

$$\max_{n\neq m,n\in\mathcal{A}_G} p_n^{\delta_{q,k_r}^{(n)}} q_n^{\delta_{j,k_c}^{(n)}} \gtrsim \max\Big(\max_{n\neq m,n\in\mathcal{A}_G} p_n^{\delta_{q,1}^{(n)}} q_n^{\delta_{j,1}^{(n)}}, \max_{n\notin\mathcal{A}_G} p_n^{\delta_{r,1}^{(n)}} q_n^{\delta_{c,1}^{(n)}}\Big) R(T, d_{fac,\star}),$$

we have

$$S_{G,m} \gtrsim_P \frac{\max_{n\neq m,n\in\mathcal{A}_G} p_n^{\delta_{q,k_r}^{(n)}} q_n^{\delta_{j,k_c}^{(n)}}}{p_m^{\delta_{q,1}^{(m)}} q_m^{\delta_{j,1}^{(m)}}}, \quad \textit{and}$$

$$S_{G,k} \lesssim_P \frac{\max\big(\max_{n\in\mathcal{A}_G} p_n^{\delta_{q,1}^{(n)}} q_n^{\delta_{j,1}^{(n)}}, \max_{n\neq k, n\notin\mathcal{A}_G} p_n^{\delta_{r,1}^{(n)}} q_n^{\delta_{c,1}^{(n)}}\big)}{p_k^{\delta_{r,1}^{(k)}} q_k^{\delta_{c,1}^{(k)}}} R(T, d_{fac,\star}),$$

where $R(\cdot,\cdot)$ *is defined in (4.1) and* $d_{fac,\star}$ *in Assumption* (R2). *In particular, if* $p_m^{\delta_{q,1}^{(m)}} q_m^{\delta_{j,1}^{(m)}} \asymp p_n^{\delta_{q,1}^{(n)}} q_n^{\delta_{j,1}^{(n)}} \asymp p_n^{\delta_{q,k_r}^{(n)}} q_n^{\delta_{j,k_c}^{(n)}}$ *for any* $n, m \in \mathcal{A}_G$, *and* $\max_{n\in[M]}(p_n^{\delta_{q,1}^{(n)}} q_n^{\delta_{j,1}^{(n)}}, p_n^{\delta_{r,1}^{(n)}} q_n^{\delta_{c,1}^{(n)}}) R(T, d_{fac,\star}) = o(\min_{k\notin\mathcal{A}_G} p_k^{\delta_{r,1}^{(k)}} q_k^{\delta_{c,1}^{(k)}})$, *then*

$$S_{G,m} \asymp_P 1, \quad S_{G,k} = o_P(1).$$

The last conditions in the above theorem translate to that all signals from the global common factors from different groups are of the same order, while the maximum global or local signals are of a smaller order than the minimum local ones after multiplying a factor of $R(T, d_{\text{fac},\star})$. In practice, we need to replace k_r, $k_r^{(m)}$, k_c, $k_c^{(m)}$ in the definition of $S_{G,m}$ by their estimators described in Section 3.4. We stress that over-estimating these numbers of factors is not much of a problem, but under-estimating them can lead to failure in detecting the existence of global common factors potentially. Below we provide a theorem for the rank estimators for all the loading matrices presented in Section 3.4.

Theorem 10 *Let all the conditions in Theorem 7 hold. Then for each* $m \in [M]$, *as* $\min(T, p_m, q_m) \to \infty$,

$$P(\widehat{r}_1 \geq r_1) \to 1, \quad P(\widehat{\ell}_1 \geq \ell_1) \to 1, \quad P(\widehat{r}_2^{(m)} \geq r_2^{(m)}) \to 1, \quad P(\widehat{\ell}_2^{(m)} \geq \ell_2^{(m)}) \to 1,$$

if $\eta_{A_{1m}} = o(\nu_{r,m}^2)$ *and* $\eta_{B_{1m}} = o(\nu_{c,m}^2)$, *where* $\eta_{A_{1m}}$ *and* $\eta_{B_{1m}}$ *are the rates defined in Theorem 3. Moreover, we also have* $P(\widehat{k}_r \geq k_r) \to 1$ *and* $P(\widehat{k}_c \geq k_c) \to 1$ *if*

$$\begin{aligned}
&\max_{m,n\in[M],n\neq m} p_m^{\delta_{q,1}^{(m)}} q_m^{\delta_{j,1}^{(m)}} p_n^{\delta_{q,1}^{(n)}} q_n^{\delta_{j,1}^{(n)}} \cdot \max_{m,n\in[M],n\neq m} \big[T^{-1} p_m q_m p_n q_n + p_m^{\delta_{q,1}^{(m)}} q_m^{\delta_{j,1}^{(m)}} p_n^{\delta_{q,1}^{(n)}} q_n^{\delta_{j,1}^{(n)}} R(T, d_{fac,\star})\big] \\
&= o\Big(\min\big(\max_{m,n\in[M],n\neq m} p_m^{2\delta_{q,k_r}^{(m)}} q_m^{2\delta_{j,1}^{(m)}} p_n^{2\delta_{q,k_r}^{(n)}} q_n^{2\delta_{j,1}^{(n)}}, \max_{m,n\in[M],n\neq m} p_m^{2\delta_{q,1}^{(m)}} q_m^{2\delta_{j,k_c}^{(m)}} p_n^{2\delta_{q,1}^{(n)}} q_n^{2\delta_{j,k_c}^{(n)}}\big)\Big).
\end{aligned}$$

Finally, we also have $P(\widehat{k}_r^{(m)} \geq k_r^{(m)}) \to 1$ *and* $P(\widehat{k}_c^{(m)} \geq k_c^{(m)}) \to 1$ *if*

$$\begin{aligned}
q_m^{2(1-\delta_{c,1}^{(m)})} + T^{-1} q_m^2 p_m + q_m^{1+\delta_{j,1}^{(m)}} p_m^{1+\delta_{q,1}^{(m)}} \eta_{Q,m} &= o(q_m^{4\delta_{c,k_c^{(m)}}^{(m)} - 2\delta_{c,1}^{(m)}} p_m^{2\delta_{r,1}^{(m)}}), \\
p_m^{2(1-\delta_{r,1}^{(m)})} + T^{-1} p_m^2 q_m + p_m^{1+\delta_{q,1}^{(m)}} q_m^{1+\delta_{j,1}^{(m)}} \eta_{J,m} &= o(p_m^{4\delta_{r,k_r^{(m)}}^{(m)} - 2\delta_{r,1}^{(m)}} q_m^{2\delta_{c,1}^{(m)}}).
\end{aligned}$$

If all factors are strong and all processes have short memory, the greater than or equal sign in all asymptotic probabilities above will become the equal sign, if we add the following terms to the numerator and denominator of the eigenvalue ratios defining $\widehat{r}_1$, $\widehat{r}_2^{(m)}$, $\widehat{\ell}_1$, $\widehat{\ell}_2^{(m)}$, $\widehat{k}_r$, $\widehat{k}_c$, $\widehat{k}_r^{(m)}$ *and* $\widehat{k}_c^{(m)}$ *respectively (*C *is a*

generic constant):

$$\max_{m\neq n\in[M]} CT^{-1}p_m p_n,\ CT^{-1}p_m^2,\ \max_{m\neq n\in[M]} CT^{-1}q_m q_n,\ CT^{-1}q_m^2$$

$$\max_{m\neq n\in[M]} CT^{-1}p_m q_m p_n q_n,\ \max_{m\neq n\in[M]} CT^{-1}p_m q_m p_n q_n,\ CT^{-1/2}p_m q_m,\ CT^{-1/2}p_m q_m.$$

The addition of the rates in the last part of Theorem 10 stabilizes the eigenvalue ratios to constant order after the dip to $o_P(1)$ at the true number of factors, hence proving the consistency of the rank estimators.

5 Practical Detection and Screening of Global Common Factors

Theorem 9 provides a theoretical grounding for detecting global common factors through the group-level statistics $S_{G,m}$, $m \in [M]$. From this, we derive two practical procedures for detecting which group of data, $\{\mathbf{Y}_t^{(m)}\}_{t\in[T]}$ for $m \in [M]$, share the global common factors. The first one investigates whether any nontrivial global common factors exist, and the second one screens which groups are driven by these factors.

For practical implementation, we standardize $\check{\mathbf{Y}}_t^{(m)}$ and define

$$\check{\mathbf{Z}}_t^{(m)} = \check{\mathbf{Y}}_t^{(m)}/\widehat{\sigma}_m, \qquad \widehat{\sigma}_m^2 = (Tp_m q_m)^{-1}\sum_{t=1}^{T}\left\|\check{\mathbf{Y}}_t^{(m)}\right\|_F^2. \tag{5.1}$$

The statistic $S_{G,m}$ in (4.3) is then computed using $\check{\mathbf{Z}}_t^{(m)}$ in place of $\check{\mathbf{Y}}_t^{(m)}$. This normalization reduces the influence of purely marginal scale differences between groups, preventing signal domination from groups just for their larger scales.

Let $S_{G,(1)} \geq S_{G,(2)} \geq \cdots \geq S_{G,(M)}$ denote the decreasing order statistics of $S_{G,1}, \ldots, S_{G,M}$. Testing the existence of global factors is equivalent to testing the following hypotheses:

$$H_{0,G} : \mathcal{A}_G = \varnothing \qquad \text{v.s.} \qquad H_{1,G} : |\mathcal{A}_G| \geq 2.$$

Theorem 9 implies that $S_{G,(1)}$ could be used for this test, since all $S_{G,m}$ decay under $H_{0,G}$, while at least two are of constant order under $H_{1,G}$. Nevertheless, we use $S_{G,(2)}$ because $H_{1,G}$ requires the global common component to be shared by at least two groups, so second-largest statistic directly reflects the alternative. Given an existence cutoff τ_1, we reject $H_{0,G}$ when $S_{G,(2)} > \tau_1$. After rejecting the null, we may further screen which groups share the global common factors. Let τ_2 be a screening cutoff and define the initial

selected set $\widehat{\mathcal{A}}_G^\circ = \{m \in [M] : S_{G,m} > \tau_2\}$. The final selected set is then

$$\widehat{\mathcal{A}}_G = \begin{cases} \widehat{\mathcal{A}}_G^\circ, & S_{G,(2)} > \tau_1 \text{ and } |\widehat{\mathcal{A}}_G^\circ| \geq 2, \\ \varnothing, & \text{otherwise.} \end{cases}$$

The definition of $\widehat{\mathcal{A}}_G$ enforces the convention that a global common component must involve at least two groups, cf. Remark 3 of Cen (2026). It also prevents reporting isolated selected groups when the evidence for a global factor is insufficient.

We next discuss how to choose the cutoffs τ_1 and τ_2. Although Theorem 9 suggests that inactive statistics decay at the rate $R(T, d_{\text{fac},\star})$, we do not directly use a deterministic rate threshold in the main procedure, since the finite-sample magnitude of $S_{G,m}$ can still depend on constants related to M, the dimensions p_m, q_m, factor strengths, among others. Instead, we use a circular-shift bootstrap to choose the existence and screening cutoffs τ_1 and τ_2.

The bootstrap is designed to preserve the within-group temporal dependence and marginal residual structure, while breaking the synchronous cross-group alignment that would be induced by global common factors. The procedure is summarized as follows.

Algorithm 1 Circular-shift bootstrap calibration for global common factor detection and screening

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1. Given the standardized arrays $\{\check{\mathbf{Z}}_t^{(m)}\}$ from (5.1), choose the bootstrap size B and calibration level α. Fix one reference group $m_0 \in [M]$.

2. For each bootstrap replication $b = 1, \ldots, B$, keep the reference group fixed: $\check{\mathbf{Z}}_t^{*(m_0,b)} = \check{\mathbf{Z}}_t^{(m_0)}$ for $t \in [T]$. For each $m \neq m_0$, draw an independent circular shift $\ell_m^{(b)}$ uniformly from $\{0, 1, \ldots, T-1\}$, and define $\check{\mathbf{Z}}_t^{*(m,b)} = \check{\mathbf{Z}}^{(m)}_{1+\{(t+\ell_m^{(b)}-1) \bmod T\}}$ for $t \in [T]$.

3. Reapply the detection procedure used to compute $S_{G,m}$ to the shifted arrays $\{\check{\mathbf{Z}}_t^{*(m,b)}\}$, and obtain $S_{G,1}^{*(b)}, \ldots, S_{G,M}^{*(b)}$.

4. For each $b \in [B]$, let $S_{G,(1)}^{*(b)} \geq S_{G,(2)}^{*(b)} \geq \cdots \geq S_{G,(M)}^{*(b)}$ denote the decreasing order statistics of $S_{G,1}^{*(b)}, \ldots, S_{G,M}^{*(b)}$. After B bootstrap replications, set

$$\tau_1 = \tau_2 = \widehat{q}_{1-\alpha}\left\{S_{G,(2)}^{*(b)} : b = 1, \ldots, B\right\},$$

where $\widehat{q}_{1-\alpha}$ denotes the empirical $(1-\alpha)$-quantile.

We use $S_{G,(2)}^{*(b)}$ to calibrate both τ_1 and τ_2, aligning the screening rule with the existence test. Comparisons in Section E.1.2 of the Supplementary Materials show that the analogous $S_{G,(1)}$-based procedure

has similar existence-test performance but is more conservative in screening and can substantially reduce recovery.

Remark 7 *The circular-shift bootstrap in Algorithm 1 is formulated under the assumption that the factor series have a single stationary segment. For data with known or unknown structural breaks, or more generally with multiple regimes, whole-sample circular shifting may mix observations from different regimes. A possible extension is to first divide the sample into regimes, either using known break points or estimated break points from change point detection methods, and then apply circular shifting within each regime.*

6 Numerical Results

6.1 Simulation

6.1.1 Data generating process

We generate a collection of matrix time series $\{\mathbf{Y}_t^{(m)}\}_{t\in[T]}$, where $\mathbf{Y}_t^{(m)} \in \mathbb{R}^{p_m \times q_m}$, from the MMEFM.

Loadings: For each group m, the entries of $\mathbf{A}_{1m}$, $\mathbf{A}_{2m}$, $\mathbf{B}_{1m}$, $\mathbf{B}_{2m}$, $\mathbf{Q}_m$, $\mathbf{J}_m$, $\mathbf{R}_m$, and $\mathbf{C}_m$ are first generated independently from $N(0,1)$. We impose the desired factor strengths by columnwise rescaling. Specifically, if $\mathbf{L}$ is a loading matrix with p rows and its jth column has strength parameter $\xi_j \in (1/2, 1]$, we set $\mathbf{L}_{\cdot,j} = p^{-(1-\xi_j)/2}\mathbf{Z}_{\cdot,j}$ with $\mathbf{Z}_{\cdot,j} \sim N(\mathbf{0}, \mathbf{I}_p)$, so that $\|\mathbf{L}_{\cdot,j}\|_2^2 \asymp p^{\xi_j}$. Each column is then demeaned to satisfy (IC1). We set $r_1 = r_2^{(m)} = \ell_1 = \ell_2^{(m)} = 2$ for the main effects, $k_r = k_c = 3$ for the global common component, and $k_r^{(m)} = k_c^{(m)} = 2$ for the local common components.

Factor dynamics: All latent factor series are generated according to the piecewise-stationary construction described in subsection 4.1. This applies to μ_t, $\mu_t^{(m)}$, and the entries of $\boldsymbol{\alpha}_t$, $\boldsymbol{\beta}_t$, $\boldsymbol{\alpha}_t^{(m)}$, $\boldsymbol{\beta}_t^{(m)}$, $\mathbf{G}_t$, and $\mathbf{F}_t^{(m)}$. We use two equal-length segments. For a generic scalar series, we first generate $x_t = \sum_{w\geq 0} \phi^w \eta_{t-w}$ with $\phi = 0.6$ and $\eta_t \sim N(0,1)$ independently over t. Within segment s, we then set $y_t = (1-B)^{-d_s} x_t$, where B denotes the backshift operator and $(1-B)^{-d_s} = \sum_{w\geq 0} \pi_w(d_s) B^w$, with $\pi_0(d_s) = 1$ and $\pi_w(d_s) = \pi_{w-1}(d_s)(w-1+d_s)/w$ for $w \geq 1$. When $d_s = 0$, this reduces to the short-memory AR(1) case. This yields segmentwise stationary dynamics with polynomially decaying dependence when $d_s > 0$, while allowing the structure to change at the midpoint. To enforce the sample orthogonality in (IC2), let $\mathbf{A} = [\boldsymbol{\alpha}_1, \ldots, \boldsymbol{\alpha}_T]'$ and $\mathbf{A}^{(m)} = [\boldsymbol{\alpha}_1^{(m)}, \ldots, \boldsymbol{\alpha}_T^{(m)}]'$. We replace $\mathbf{A}^{(m)}$ by $(\mathbf{I} - \mathbf{P}_\alpha)\mathbf{A}^{(m)}$, where $\mathbf{P}_\alpha = \mathbf{A}(\mathbf{A}'\mathbf{A})^{-1}\mathbf{A}'$. Thus $\mathbf{A}'\mathbf{A}^{(m)} = 0$. We apply the same construction to $\boldsymbol{\beta}_t$ and $\boldsymbol{\beta}_t^{(m)}$. The local grand means are demeaned across groups at each t so that $\sum_{m=1}^M \mu_t^{(m)} = 0$.

Idiosyncratic noise: For $\mathbf{Q}_{e,m}$, $\mathbf{J}_{e,m}$, $\mathbf{R}_{e,m}$, and $\mathbf{C}_{e,m}$, entries are first generated independently from $N(0,1)$ and are then set to zero independently with probability 0.95. The entries of $\mathbf{G}_{e,t}$ and $\mathbf{F}_{e,t}^{(m)}$ follow

independent AR(1) processes with coefficient 0.2 and standard normal innovations. The entries of $\boldsymbol{\epsilon}_t^{(m)}$ are generated in the same way. Finally, the entries of $\boldsymbol{\Sigma}_{\epsilon,m}$ are generated independently from $|N(0,1)|$.

Unless otherwise stated, we take $M = 3$ groups and use 500 Monte Carlo replications in each setting. To evaluate the accuracy of our estimators for time-varying quantities, given any series of parameters $\boldsymbol{\theta} = \{\boldsymbol{\theta}_t\}_{t\in[T]}$, where $\boldsymbol{\theta}_t$ can be a scalar, vector, or matrix, and its estimator $\widehat{\boldsymbol{\theta}} = \{\widehat{\boldsymbol{\theta}}_t\}_{t\in[T]}$, we define the (relative) mean squared error as

$$\mathrm{MSE}(\widehat{\boldsymbol{\theta}}) := \frac{\sum_{t=1}^{T} \|\boldsymbol{\theta}_t - \widehat{\boldsymbol{\theta}}_t\|_F^2}{\sum_{t=1}^{T} \|\boldsymbol{\theta}_t\|_F^2}.$$

Moreover, for any given loading matrix $\mathbf{Q}$ and its estimator $\widehat{\mathbf{Q}}$, we use the column-space distance

$$\mathcal{D}(\mathbf{Q}, \widehat{\mathbf{Q}}) := \left\| \mathbf{Q}(\mathbf{Q}'\mathbf{Q})^{-1}\mathbf{Q}' - \widehat{\mathbf{Q}}(\widehat{\mathbf{Q}}'\widehat{\mathbf{Q}})^{-1}\widehat{\mathbf{Q}}' \right\|,$$

which is widely used in the literature such as Chen et al. (2022), among others. For group-specific quantities, MSE or $\mathcal{D}$ is first computed within each group and then averaged over m.

6.1.2 Accuracy of main effects

We first study estimation of the grand means and main effects under four designs (Ia)–(Id). Each design is considered at $T \in \{100, 400\}$. The common component part of the DGP follows the baseline design (IIa) in subsection 6.1.3.

(Ia) **Main effect baseline.** Set $(p_m, q_m) = (30, 30)$ for each $m \in [M]$. All loading strength parameters in (ME1) are set to 1. For all latent series, we set memory parameters $d_1 = d_2 = 0$.

(Ib) **Long-memory main effects.** Same as (Ia), except that $d_1 = d_2 = 0.1$ for μ_t, $\mu_t^{(m)}$, and each entry of $\boldsymbol{\alpha}_t$, $\boldsymbol{\beta}_t$, $\boldsymbol{\alpha}_t^{(m)}$, and $\boldsymbol{\beta}_t^{(m)}$.

(Ic) **Weaker main effect factors.** Same as (Ia), except that $\gamma_r = \gamma_c = \gamma_r^{(m)} = \gamma_c^{(m)} = 0.8$ for each $m \in [M]$.

(Id) **Larger imbalanced dimensions.** Same as (Ia), except that $(p_m, q_m) = (30, 60)$ for each $m \in [M]$.

Table 1 reports the results. The direct moment estimators change little with T, since their pointwise accuracy is driven by cross-sectional averaging; the slightly smaller relative MSEs under long memory in (Ib) mainly reflect the larger signal magnitude in the denominator rather than improved estimation accuracy. In contrast, the main effect loading space errors decrease clearly with T, consistent with Theorems 3–5. Long memory mainly increases dispersion, while weaker loadings moderately reduce accuracy. Increasing only q_m in (Id) gives a clearer improvement on the column side. The initial and refitted global loading estimates have

nearly identical errors, indicating that refitting mainly links the loadings to their corresponding estimated global main effect factor series rather than changing their estimated spaces. Accordingly, the global main effect products improve sharply after refitting, while the local products also become more accurate as T increases. Long memory and weaker loadings mainly affect small-T accuracy, while the imbalanced design remains stable.

Table 1: Main effect estimation accuracy. All entries are Monte Carlo means multiplied by 10^2, with standard deviations in parentheses.

Settings	Ia		Ib		Ic		Id	
	$T = 100$	$T = 400$	$T = 100$	$T = 400$	$T = 100$	$T = 400$	$T = 100$	$T = 400$
MSE($\widehat{c}_t^{(m)}$)	0.05(0.01)	0.05(0.00)	0.04(0.01)	0.04(0.00)	0.05(0.01)	0.05(0.00)	0.02(0.00)	0.02(0.00)
MSE($\widehat{\mathbf{a}}_t^{(m)}$)	0.59(0.07)	0.58(0.06)	0.48(0.07)	0.47(0.04)	1.17(0.13)	1.14(0.11)	0.30(0.04)	0.29(0.03)
MSE($\widehat{\mathbf{b}}_t^{(m)}$)	0.60(0.07)	0.58(0.05)	0.49(0.07)	0.47(0.05)	1.17(0.14)	1.16(0.11)	0.59(0.06)	0.58(0.04)
MSE($\widehat{\mu}_t$)	0.03(0.01)	0.03(0.00)	0.02(0.01)	0.02(0.00)	0.03(0.01)	0.03(0.00)	0.01(0.00)	0.01(0.00)
MSE($\widehat{\mu}_t^{(m)}$)	0.08(0.01)	0.08(0.01)	0.06(0.01)	0.06(0.01)	0.08(0.02)	0.08(0.01)	0.04(0.01)	0.04(0.00)
$\mathcal{D}(\mathbf{A}_{1m}, \widehat{\mathbf{A}}_{1m})$	6.53(4.16)	2.77(1.11)	8.27(7.02)	3.19(1.63)	6.82(4.71)	2.89(1.07)	6.14(4.46)	2.57(1.02)
$\mathcal{D}(\mathbf{A}_{1m}, \widetilde{\mathbf{A}}_{1m})$	6.51(4.16)	2.76(1.11)	8.26(7.02)	3.19(1.63)	6.79(4.71)	2.88(1.07)	6.14(4.46)	2.57(1.02)
$\mathcal{D}(\mathbf{A}_{2m}, \widehat{\mathbf{A}}_{2m})$	5.91(2.37)	2.72(0.86)	7.55(4.45)	3.12(1.21)	6.43(2.71)	2.89(0.91)	5.69(2.72)	2.57(0.88)
$\mathcal{D}(\mathbf{B}_{1m}, \widehat{\mathbf{B}}_{1m})$	6.78(5.27)	2.76(1.07)	9.64(9.27)	3.26(1.46)	7.07(4.26)	2.82(0.95)	4.92(4.10)	2.01(0.71)
$\mathcal{D}(\mathbf{B}_{1m}, \widetilde{\mathbf{B}}_{1m})$	6.76(5.27)	2.76(1.07)	9.62(9.27)	3.26(1.46)	7.05(4.26)	2.80(0.95)	4.91(4.10)	2.00(0.71)
$\mathcal{D}(\mathbf{B}_{2m}, \widehat{\mathbf{B}}_{2m})$	6.28(3.54)	2.68(0.86)	8.38(6.53)	3.16(1.15)	6.59(2.57)	2.82(0.78)	4.57(3.22)	1.98(0.59)
MSE($\widehat{\mathbf{A}}_{1m}\widehat{\boldsymbol{\alpha}}_t$)	101.74(10.48)	102.27(10.74)	102.17(9.27)	101.31(8.93)	104.18(14.56)	104.66(14.85)	102.22(9.85)	102.09(10.05)
MSE($\widetilde{\mathbf{A}}_{1m}\widehat{\boldsymbol{\alpha}}_t$)	0.44(0.60)	0.10(0.06)	0.89(2.41)	0.12(0.13)	0.60(1.85)	0.13(0.07)	0.45(1.99)	0.07(0.05)
MSE($\widehat{\mathbf{A}}_{2m}\widehat{\boldsymbol{\alpha}}_t^{(m)}$)	0.59(0.58)	0.18(0.07)	1.11(2.49)	0.20(0.13)	0.82(1.59)	0.28(0.08)	0.56(1.46)	0.13(0.06)
MSE($\widehat{\mathbf{B}}_{1m}\widehat{\boldsymbol{\beta}}_t$)	101.86(9.91)	102.09(10.26)	102.14(10.31)	101.64(9.40)	103.32(15.62)	104.52(15.73)	101.31(7.17)	101.09(7.34)
MSE($\widetilde{\mathbf{B}}_{1m}\widehat{\boldsymbol{\beta}}_t$)	0.62(2.25)	0.10(0.05)	1.41(4.77)	0.12(0.10)	0.56(0.90)	0.13(0.06)	0.31(1.30)	0.05(0.03)
MSE($\widehat{\mathbf{B}}_{2m}\widehat{\boldsymbol{\beta}}_t^{(m)}$)	0.76(1.97)	0.18(0.07)	1.60(5.09)	0.21(0.11)	0.78(0.80)	0.27(0.07)	0.43(2.26)	0.09(0.03)

6.1.3 Accuracy of common components

We next study estimation of the global and local common components under four designs (IIa)–(IId). Each design is considered at $T \in \{100, 400\}$. The main effect part of the DGP follows the baseline design (Ia) in subsection 6.1.2.

(IIa) **Common component baseline.** Set $(p_m, q_m) = (30, 30)$ for each $m \in [M]$. All loading strength parameters in (L1) are set to 1. For all latent series, we set memory parameters $d_1 = d_2 = 0$.

(IIb) **Long-memory common factors.** Same as (IIa), except that $d_1 = d_2 = 0.1$ for each entry of $\mathbf{G}_t$ and $\mathbf{F}_t^{(m)}$.

(IIc) **Weaker common factors.** Same as (IIa), except that for each $m \in [M]$, $\delta_{q,i}^{(m)} = \delta_{j,i}^{(m)} = 0.8$ for $i = 1, 2, 3$ and $\delta_{r,i}^{(m)} = \delta_{c,i}^{(m)} = 0.8$ for $i = 1, 2$.

(IId) **Larger imbalanced dimensions.** Same as (IIa), except that $(p_m, q_m) = (30, 60)$ for each $m \in [M]$.

Table 2 reports the results. The refitted global loading estimates $\widetilde{\mathbf{Q}}_m$ and $\widetilde{\mathbf{J}}_m$ are substantially more accurate than the initial estimates $\widehat{\mathbf{Q}}_m$ and $\widehat{\mathbf{J}}_m$ in every design, showing the benefit of cross-group alignment

and refitting. The loading-space errors also fall clearly as T increases; weaker loadings in (IIc) make estimation more difficult, while the larger imbalanced dimensions in (IId) remain stable and generally improve accuracy. The improvement is even more pronounced for the global common component product, whose error drops sharply after alignment and refitting; the local product also improves with T. Under long memory, several $T = 100$ entries have large Monte Carlo variation, but this largely disappears at $T = 400$.

Table 2: Common component estimation accuracy. All entries are Monte Carlo means multiplied by 10^2, with standard deviations in parentheses.

Settings	IIa		IIb		IIc		IId	
	$T = 100$	$T = 400$	$T = 100$	$T = 400$	$T = 100$	$T = 400$	$T = 100$	$T = 400$
$\mathcal{D}(\mathbf{Q}_m, \widehat{\mathbf{Q}}_m)$	7.98(3.09)	3.90(1.45)	9.84(5.71)	4.52(1.77)	9.21(3.03)	4.46(1.41)	5.75(2.21)	2.73(1.05)
$\mathcal{D}(\mathbf{Q}_m, \widetilde{\mathbf{Q}}_m)$	1.77(0.40)	0.79(0.15)	2.13(4.02)	0.79(0.18)	3.17(0.46)	1.44(0.18)	1.13(0.20)	0.50(0.07)
$\mathcal{D}(\mathbf{J}_m, \widehat{\mathbf{J}}_m)$	8.22(3.15)	3.93(1.57)	9.69(5.65)	4.51(1.74)	9.27(3.16)	4.24(1.25)	8.24(3.12)	3.73(1.32)
$\mathcal{D}(\mathbf{J}_m, \widetilde{\mathbf{J}}_m)$	1.56(0.20)	0.71(0.08)	1.75(3.54)	0.67(0.08)	3.01(0.37)	1.38(0.16)	1.46(0.15)	0.66(0.06)
$\mathcal{D}(\mathbf{R}_m, \widehat{\mathbf{R}}_m)$	2.53(0.64)	1.01(0.18)	3.28(4.47)	1.05(0.22)	4.23(0.79)	2.42(0.55)	2.23(0.72)	0.76(0.15)
$\mathcal{D}(\mathbf{C}_m, \widehat{\mathbf{C}}_m)$	2.50(0.66)	1.01(0.16)	3.33(4.54)	1.04(0.22)	4.23(0.90)	2.41(0.56)	2.09(0.48)	0.84(0.10)
MSE($\widehat{\mathbf{Q}}_m \widehat{\mathbf{G}}_t \widehat{\mathbf{J}}'_m$)	155.20(39.46)	153.60(34.79)	155.72(38.31)	155.31(36.50)	156.11(38.47)	156.13(34.55)	152.32(37.03)	153.54(36.18)
MSE($\widetilde{\mathbf{Q}}_m \widehat{\mathbf{G}}_t \widetilde{\mathbf{J}}'_m$)	0.07(0.02)	0.04(0.01)	0.49(9.09)	0.03(0.01)	0.25(0.05)	0.16(0.02)	0.04(0.01)	0.02(0.00)
MSE($\widehat{\mathbf{R}}_m \widehat{\mathbf{F}}_t^{(m)} \widehat{\mathbf{C}}'_m$)	0.25(0.11)	0.13(0.03)	0.82(10.24)	0.12(0.03)	0.71(0.20)	0.48(0.10)	0.14(0.06)	0.06(0.01)

To assess our common component estimation procedure, we compare it with the basic and augmented MFM estimators of Zhang et al. (2025). For comparability, all three estimators use the same $\breve{\mathbf{Y}}_t^{(m)}$ for common component estimation. In addition to setting (IIa), we consider setting (IIc') which is the same as (IIa), except that the global loading strengths are $(1, 0.8, 0.8)$ in both modes, while the local loading strengths remain $(1, 1)$. The row and column designs are symmetric, so we report only the row-side results. Table 3 shows that the augmented MFM improves substantially on the basic estimator, while our estimator yields smaller errors throughout. The advantage is more pronounced when the global loadings are weaker.

Table 3: Global row loading-space estimation accuracy for MMEFM and MFM estimators. MFM Augmented uses 15 iterations. All entries are Monte Carlo means multiplied by 10^2, with standard deviations in parentheses.

Settings	IIa		IIc′	
	$T = 100$	$T = 400$	$T = 100$	$T = 400$
MMEFM	1.78(0.42)	0.80(0.14)	5.15(13.91)	1.30(0.22)
MFM Basic	14.96(8.95)	3.84(1.34)	69.33(21.23)	17.08(11.93)
MFM Augmented	2.71(2.08)	1.22(0.16)	10.07(13.33)	2.05(0.30)

6.1.4 Behavior of the statistic $S_{G,m}$

We next examine the finite-sample behavior of the statistic $S_{G,m}$ in Theorem 9. Let $\mathcal{A}_G$ denote the active set of groups containing a shared global common component, and let $\mathcal{A}_G^c = [M] \setminus \mathcal{A}_G$. Since a global common component must be shared by at least two groups, we consider $|\mathcal{A}_G| \in \{0, 2, 3, 4, 5\}$ with $M = 5$.

The active groups are randomly selected in each Monte Carlo replication. We use 400 replications and set $T = 200$ unless otherwise stated.

We consider six settings. The first four settings are inherited from (IIa)–(IId) and are relabelled as (IIIa)–(IIId), respectively. We additionally consider

(IIIe) **Weaker local common component.** Same as (IIIa), except that $\delta_{r,i}^{(m)} = \delta_{c,i}^{(m)} = 0.8$ for $i = 1, 2$ and each $m \in [M]$.

(IIIf) **Weaker global common component.** Same as (IIIa), except that $\delta_{q,i}^{(m)} = \delta_{j,i}^{(m)} = 0.8$ for $i = 1, 2, 3$ and each $m \in [M]$.

Before computing $S_{G,m}$, we standardize $\check{\mathbf{Y}}_t^{(m)}$ using (5.1). Figure 1 reports the within-replication averages of $S_{G,m}$ over the active and inactive groups at $T = 200$. The active and inactive averages are clearly separated in settings (IIIa)–(IIIe). The weaker-global setting (IIIf) is more difficult, especially when only a few groups are active. The separation also tends to increase with $|\mathcal{A}_G|$, as each active group then shares the global component with more groups.

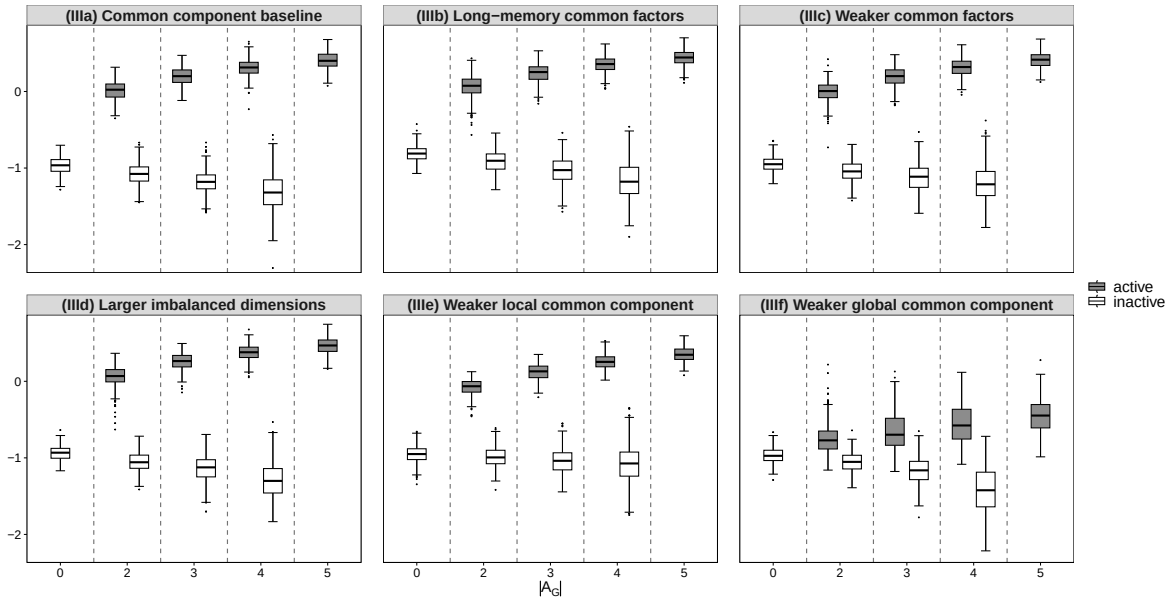


Figure 1: Averages of $S_{G,m}$ over the active and inactive groups at $T = 200$. The y-axis is $\log_{10}$-transformed.

Figure 2 plots the Monte Carlo medians of these active and inactive averages as T increases. Across all settings, the active averages remain relatively stable, while the inactive averages decrease. The gap therefore widens with T, consistent with Theorem 9. In the weaker-global setting (IIIf), the active averages are lower in finite samples but remain much more stable than the inactive averages.

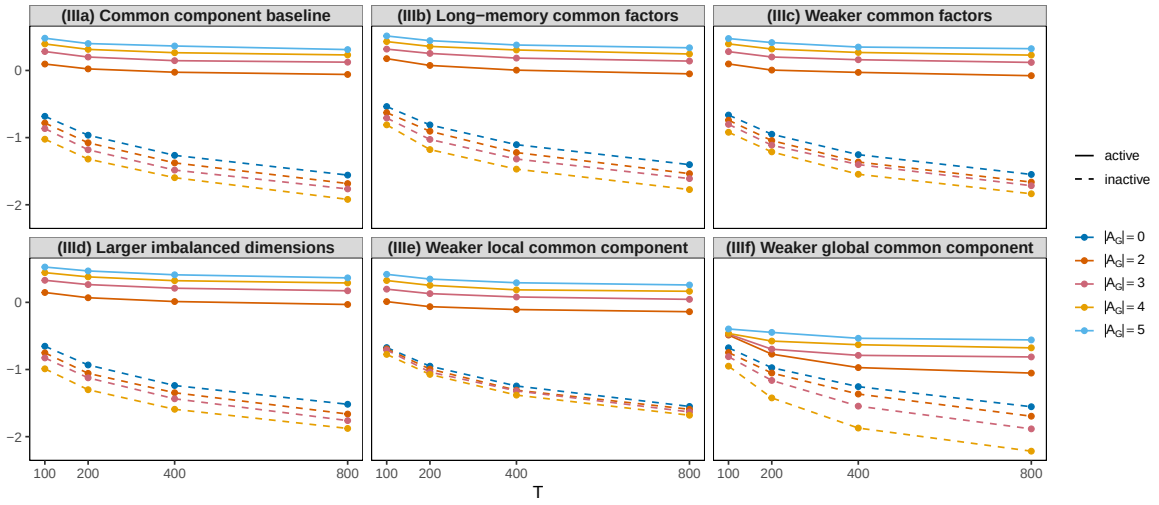


Figure 2: Medians of the averages of $S_{G,m}$ over the active and inactive groups, respectively. The y-axis is $\log_{10}$-transformed. Solid and dashed lines correspond to active and inactive groups, respectively.

6.1.5 Accuracy of detection and screening

We next examine whether the separation of $S_{G,m}$ translates into reliable detection and screening using the procedure in Section 5. For these experiments, we use 400 Monte Carlo replications and true factor ranks, set the nominal level to 5%, and use $B = 199$ bootstrap replications.

Tables 4 and 5 report the performance of the bootstrap existence test. In the baseline setting, the empirical size remains close to the nominal 5% level across sample sizes, while power increases rapidly with both T and $|\mathcal{A}_G|$. At $T = 200$, the test continues to control size well across settings. Power remains high in settings (IIIa)–(IIIe), with some loss under long memory when only a few groups are active. The weaker-global setting (IIIf) is substantially more difficult, consistent with the lower active statistics observed in Section 6.1.4. Power nevertheless improves as more groups share the global component.

Table 4: Bootstrap existence test in the baseline setting (IIIa). Entries are empirical rejection rates with Monte Carlo standard errors in parentheses; $|\mathcal{A}_G| = 0$ reports empirical size and $|\mathcal{A}_G| \geq 2$ reports power.

T	$\lvert\mathcal{A}_G\rvert = 0$	$\lvert\mathcal{A}_G\rvert = 2$	$\lvert\mathcal{A}_G\rvert = 3$	$\lvert\mathcal{A}_G\rvert = 4$	$\lvert\mathcal{A}_G\rvert = 5$
100	0.043 (0.010)	0.828 (0.019)	0.993 (0.004)	1.000 (0.000)	1.000 (0.000)
200	0.058 (0.012)	0.995 (0.004)	1.000 (0.000)	1.000 (0.000)	1.000 (0.000)
400	0.050 (0.011)	0.998 (0.002)	1.000 (0.000)	1.000 (0.000)	1.000 (0.000)

Table 5: Bootstrap existence test across settings at $T = 200$. Entries are described in Table 4.

$\lvert\mathcal{A}_G\rvert$	(IIIa)	(IIIb)	(IIIc)	(IIId)	(IIIe)	(IIIf)
0	0.058 (0.012)	0.068 (0.013)	0.050 (0.011)	0.055 (0.011)	0.058 (0.012)	0.053 (0.011)
2	0.995 (0.004)	0.930 (0.013)	0.993 (0.004)	0.985 (0.006)	1.000 (0.000)	0.085 (0.014)
4	1.000 (0.000)	1.000 (0.000)	1.000 (0.000)	1.000 (0.000)	1.000 (0.000)	0.648 (0.024)

Tables 6 and 7 report the screening performance, where TPR $= |\widehat{\mathcal{A}}_G \cap \mathcal{A}_G|/|\mathcal{A}_G|$, FPR $= |\widehat{\mathcal{A}}_G \cap$

$\mathcal{A}_G^c|/|\mathcal{A}_G^c|$, and Exact denotes $\mathbb{1}\{\widehat{\mathcal{A}}_G = \mathcal{A}_G\}$, averaged over replications. In the baseline setting, screening becomes nearly exact as T increases when global common factors are present, while false positives remain rare. The screening procedure therefore becomes increasingly accurate as the separation in $S_{G,m}$ strengthens. At $T = 200$, screening remains accurate across settings (IIIa)–(IIIe), with high true positive rates and low false positive rates. The weaker-global setting (IIIf) is again the most challenging, where the reduced shared signal leads to substantially lower true positive and exact-selection rates. This pattern closely follows the existence-test results, since the final selected set is gated by the existence decision.

Table 6: Bootstrap screening in the baseline setting (IIIa). Entries are empirical averages computed from the final gated selected set $\widehat{\mathcal{A}}_G$.

Metric	$\lvert\mathcal{A}_G\rvert = 0$			$\lvert\mathcal{A}_G\rvert = 2$			$\lvert\mathcal{A}_G\rvert = 3$			$\lvert\mathcal{A}_G\rvert = 4$			$\lvert\mathcal{A}_G\rvert = 5$		
	100	200	400	100	200	400	100	200	400	100	200	400	100	200	400
TPR	–	–	–	0.826	0.995	0.998	0.951	0.998	1.000	0.971	0.996	0.999	0.985	0.995	1.000
FPR	0.019	0.026	0.021	0.001	0.002	0.000	0.000	0.000	0.000	0.000	0.000	0.000	–	–	–
Exact	0.958	0.943	0.950	0.825	0.990	0.998	0.868	0.993	1.000	0.888	0.983	0.998	0.928	0.975	1.000

Table 7: Bootstrap screening across settings at $T = 200$. Entries are described in Table 6.

$\lvert\mathcal{A}_G\rvert$	Metric	(IIIa)	(IIIb)	(IIIc)	(IIId)	(IIIe)	(IIIf)
0	FPR	0.026	0.029	0.023	0.024	0.025	0.022
	Exact	0.943	0.933	0.950	0.945	0.943	0.948
2	TPR	0.995	0.930	0.993	0.985	1.000	0.073
	FPR	0.002	0.000	0.001	0.002	0.029	0.009
	Exact	0.990	0.930	0.990	0.980	0.915	0.060
4	TPR	0.996	0.986	0.995	0.999	0.999	0.425
	FPR	0.000	0.000	0.000	0.000	0.005	0.003
	Exact	0.983	0.950	0.980	0.998	0.990	0.085

Finally, as a robustness check, we repeat the estimation experiments and the baseline statistic-separation experiment under Student $t(6)$ innovations, with the corresponding results reported in Section E.1.1 of the Supplementary Materials. The qualitative patterns remain largely unchanged. Heavy tails mainly increase finite-sample variation, especially for smaller T, while having little effect on the overall estimation accuracy or the separation between active and inactive groups.

6.2 Real data analysis

We study daily U.S. equity data from CRSP, grouping stocks into the 11 GICS sectors. The set of stocks is fixed on December 30, 2022 by selecting the 30 largest U.S. companies by market capitalization within each sector. Six stocks with structural exits during 2023–2024 are excluded from the primary sample, leaving 324 stocks in total. We consider nine indicators: daily return, 20-day cumulative log return, log dollar volume, log turnover, 20-day relative turnover, 20-day log realized volatility, high-low range, log bid-ask spread,

and 20-day log Amihud illiquidity (Amihud, 2002). The sample runs from January 3, 2023 to December 31, 2024, with $T = 502$ trading days, giving 11 grouped matrix time series $\{\mathbf{Y}_t^{(m)}\}_{t=1}^T$. Observations in 2025 are reserved for out-of-sample forecasting.

For each indicator, the mean and standard deviation are computed from all observed entries in the 2023–2024 primary sample before imputation and used for standardization. Ordinary missing values are imputed using same-day sector medians, and each series is then demeaned over 2023–2024. We continue to write $\{\mathbf{Y}_t^{(m)}\}$ for the resulting cell-demeaned standardized matrices. Indicator construction and preprocessing are described in subsection E.2.1 of the Supplementary Materials.

Why a MMEFM structure? We first motivate the two features of the MMEFM that matter here: explicit main effects and a shared global structure.

First, under (IC1), we have the decomposition $\|\mathbf{Y}_t^{(m)}\|_F^2 = p_m q_m (\widehat{c}_t^{(m)})^2 + q_m \|\widehat{\mathbf{a}}_t^{(m)}\|^2 + p_m \|\widehat{\mathbf{b}}_t^{(m)}\|^2 + \|\check{\mathbf{Y}}_t^{(m)}\|_F^2$. The interaction component accounts only for $\sum_m \sum_t \|\check{\mathbf{Y}}_t^{(m)}\|_F^2 / \sum_m \sum_t \|\mathbf{Y}_t^{(m)}\|_F^2 \approx 56.77\%$ of the total squared magnitude, leaving 43.23% attributable to the additive components. This sizable contribution motivates modeling the main effects explicitly.

Second, we apply the global-common-factor detection and screening procedure developed in Section 5. The two largest statistics are $S_{G,(1)} = 8.254$ and $S_{G,(2)} = 7.571$, both exceeding the 95% circular-shift bootstrap cutoff of 4.589, providing evidence for a nontrivial global common component. Sector-level screening retains Industrials, Communication Services, and Real Estate. We nevertheless estimate the MMEFM using all 11 sectors, since the remaining $S_{G,m}$'s are also away from zero and may reflect weaker shared market dynamics. For comparison, we also consider a screened specification in which the global common component is estimated using only the three retained sectors, while local components are retained for all sectors. Further evidence on the contributions of the main effects and global component, together with the complete set of sector-level $S_{G,m}$ statistics, is provided in the Supplementary Materials.

Model estimation and interpretation. We next estimate the MMEFM on our stock data. The factor dimensions are selected using the procedure in subsection 3.4. The global main-effect and common-component ranks are both $(1, 1)$, while the local ranks vary across sectors, with the complete selection provided in Supplementary Table E.6.

We first examine the global structure. Figure 3 shows that the global indicator-main loadings $\widetilde{\mathbf{B}}_{1m}$ are broadly similar across sectors, with turnover and relative turnover loading in the opposite direction to daily and 20-day returns, suggesting a market-wide contrast between trading activity and recent price performance. Figure 4 further shows that the global common component has more heterogeneous indicator-loading profiles across the three selected sectors: 20-day return is most prominent in Industrials, whereas

liquidity and trading measures are more prominent in Communication Services and Real Estate. The shared global factor also shows a large movement around the late-July–early-August 2024 market dislocation, pointing to a market-wide change rather than a sector-specific episode.

We next illustrate the local structure using Financials and Information Technology. Figure 5 shows that the Financials stock-main loadings broadly separate large banks and broker-dealers from insurers and several market-infrastructure firms. Figure 6 shows that the Information Technology common stock loadings distinguish semiconductor and chip-equipment firms from software, payment, and other large technology firms, while returns load in the opposite direction to trading activity and the high-low range.

Additional estimates are reported in the Supplementary Materials.

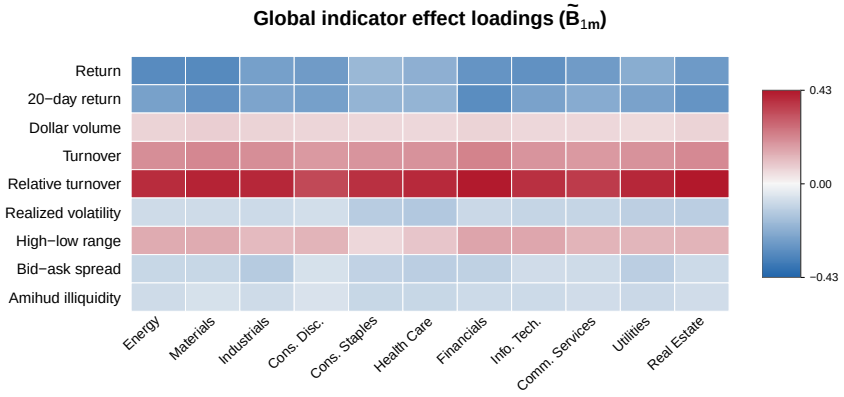


Figure 3: Global indicator main-effect estimates.

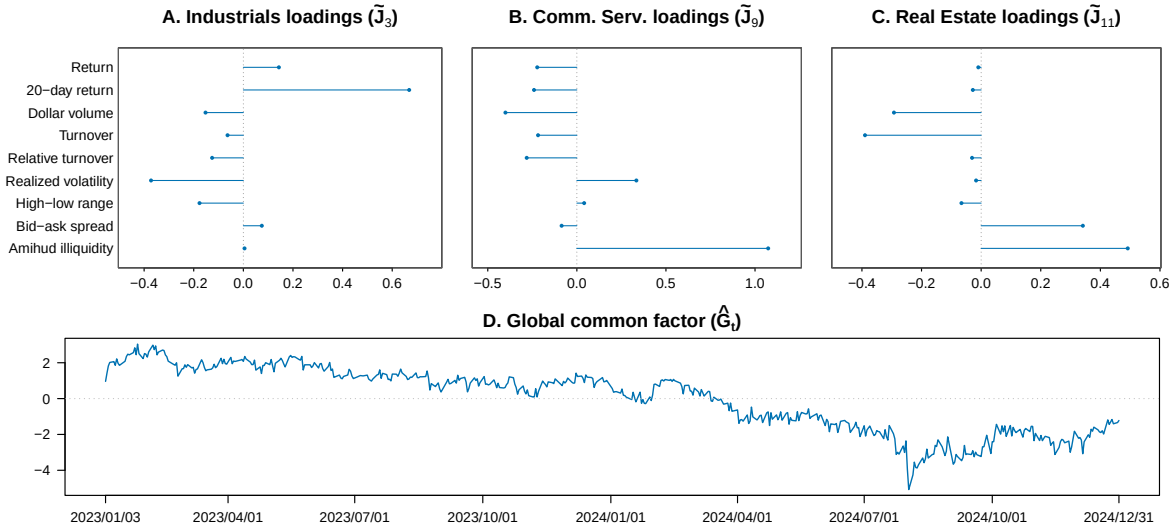


Figure 4: Global common component estimates.

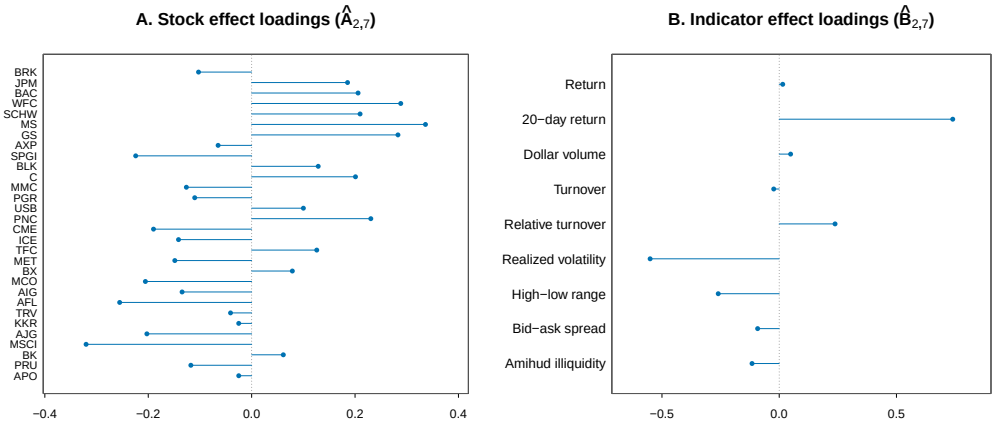


Figure 5: Local main-effect estimates for Financials.

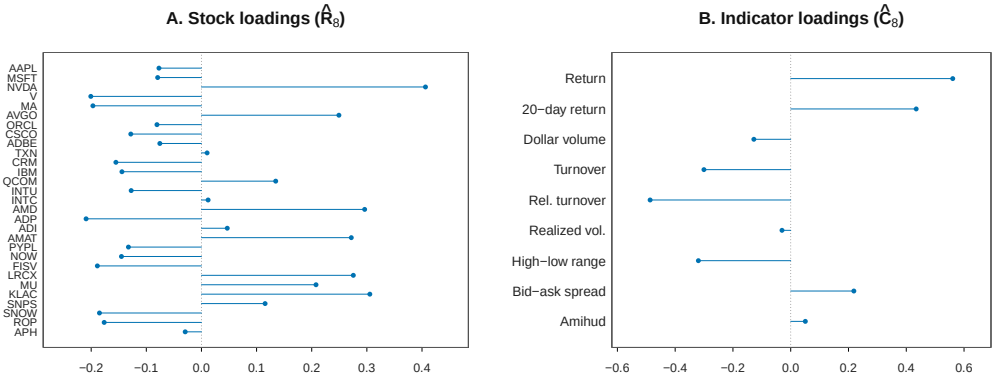


Figure 6: Local common-component estimates for Information Technology.

Model comparison. We next compare the in-sample fit of several MMEFM specifications and MFM. Table 8 reports RSS/TSS together with a simple parameter-count adjustment, $(RSS/TSS) \cdot N/(N-\nu)$, where $N := \sum_{m=1}^{M} Tp_m q_m$ and $\nu := T \cdot \#\{\text{latent series per date}\} + \#\{\text{loading coefficients}\}$.

The local-only and global-only specifications retain only local and global dynamics, respectively. MFM Basic uses the baseline estimator of Zhang et al. (2025), whereas MFM Augmented uses their Section 3.5 iterative estimator with 15 iterations. The MMEFM specifications fit better than the two single-level structured benchmarks, while the screened-global specification achieves essentially the same fit as the all-sector model with fewer loading coefficients. MFM Augmented improves markedly over MFM Basic, but both fit less well than the MMEFM specifications.

We complement the comparison with daily rolling forecasts for 2025. At each origin, ranks and loading matrices are re-estimated on a 252-day window advanced by one trading day; the recovered time-varying quantities are forecast entrywise by AR(p) models with $p \in [5]$ chosen by BIC from the latest 250 observations. We consider $h = 1, 3, 5, 10$, keep the three screened sectors fixed throughout 2025, and define MSPE

Table 8: In-sample reconstruction across MMEFM specifications and MFM benchmarks. The MFM rank estimator selects global rank $(1, 1)$; the local ranks are $(3, 3)$ for Energy, $(1, 2)$ for Information Technology, $(2, 3)$ for Consumer Discretionary, Consumer Staples, and Communication Services, and $(1, 3)$ otherwise.

Model	Per-date dim.	Loading coeffs.	RSS/TSS	Adjusted
MMEFM, all sectors	68	2,074	0.712	0.730
MMEFM, screened global	68	1,737	**0.708**	**0.726**
MMEFM, local only	64	1,228	0.742	0.760
MMEFM, global only	4	846	0.869	0.871
MFM Basic	58	1,180	1.001	1.022
MFM Augmented	58	1,180	0.771	0.787

as the total squared Frobenius forecast error divided by the number of evaluated entries.

Table 9 shows that both MMEFM specifications outperform MFM Basic at every horizon. Relative to MFM Augmented, the all-sector MMEFM reduces MSPE by around 10%, and the screened-global specification performs similarly.

Table 9: Daily rolling forecasting MSPEs for 2025.

Model	$h = 1$	$h = 3$	$h = 5$	$h = 10$
MMEFM, all sectors	**0.802**	**0.870**	**0.894**	**0.940**
MMEFM, screened global	0.810	0.875	0.897	**0.940**
MFM, Basic	0.983	1.022	1.028	1.038
MFM, Augmented	0.908	0.953	0.973	1.003

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